Canard-Root Transfer in Finite Retarded Networks and a Conditional Framework for Physical Pulse Thresholds

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Research draft: the preferred-scale selected two-return domain, six numerical Hessian blocks, effective stable-sheet radius and slope, periodic-orbit stable-set identification, and two-sided routing gates identified below are not yet closed; no biological-onset or safety-coordinate theorem is claimed.

Abstract

We separate two codimension-one objects in delayed excitable networks: a phase-fixed complete-history canard root and a physical pulse threshold given by intersection with the stable sheet of an unstable periodic orbit. For finite nonnegative Dobrushin networks, we prove network-size-independent transverse decay and a causal complete-line Green bound. An exact row reduction and corrected full cokernel transfer any supplied scalar simple root through nonbalanced delayed layers. A dimension-uniform one-gap theorem gives an explicit root radius and quadratic remainder, and a heterogeneous shared-resource family has a nonzero uniform response without a synchrony quotient.

For finite-dimensional fixed-delay C^k RFDEs, a common event window beginning after k maximum delays yields a C^k complete-history hit under tube, endpoint-sign and positive-speed hypotheses. In a two-delay leaky FitzHugh–Nagumo RFDE, an orbit-aware weighted-energy estimate proves a common C^2 near-two-period selected hit for arbitrary continuous reduced histories on the explicit scale 1.2×10^{-8}, improving a scalar comparison by about 22 decimal orders. Its center derivative is A^2 without a nonlinear one-period map; the two-step stable and unstable-inverse rates are at most 0.01 and 0.302184. Finite sampling cannot recover the required Hessian uniformly on a C^0 ball. An exact stable-row quotient and split-projective signed-bimeasure residual theorem provide the appropriate continuous-history norm, while hyperbolicity gives a qualitative local C^2 stable graph. No effective graph radius or slope is obtained, and numerical ingress into the six Hessian caps remains zero.

For a physical pulse interval, we validate a common event, its event-aligned derivative, opposite endpoint functional signs and stable-coordinate containment. A preferred-scale stable graph, its identification with the periodic orbit's stable-set germ, and its pulse crossing remain open. Biological onset and frequency–amplitude–safety control additionally require outer capture, two-sided routing, parameter-uniform response and asynchronous network safety.

Keywords. canard connection; delay differential equation; finite network; Dobrushin contraction; physical pulse threshold; frequency–amplitude control

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1 Introduction

Canard dynamics turn a small parameter displacement into a large change of orbit geometry. In ordinary slow–fast systems this mechanism is organized locally near a loss of normal hyperbolicity

(Krupa and Szmolyan, 2001a,b; Wechselberger, 2012). Delay equations add a more basic ambiguity: the state is a history, and a current-state projection can discard precisely the transverse delay response that later returns to the critical direction. Delayed canards and delayed FitzHugh–Nagumo oscillations are by now well established (Campbell et al., 2009; Krupa and Touboul, 2016a,b), but a local canard calculation does not automatically survive embedding into a large retarded network.

A second threshold problem appears when the system is actuated. A finite physical pulse generates a curve in the RFDE history space. A selected pulse/stable-sheet threshold is a value at which that curve meets a declared codimension-one stable sheet. It becomes a biological onset only after the two sides are proved to route to different attracting objects. Neither notion is the same as a Lin-gap canard root, and a projected observable event is not by itself any of the three. Our organizing principle is therefore to keep these objects separate while exposing the transverse mechanism common to them.

1.1 Three levels of generality

The paper distinguishes three statements.

First, canonical synchrony gives an exact upper-triangular block decomposition, and the transverse complete inverse gives a bounded codomain row reduction to the scalar/transverse direct sum. When all endpoint, phase, parameter, and gap data are collective, the corrected full cokernel transfers the scalar Fredholm data and gap without changing the root.

Second, a one-gap structural theorem allows small network residuals that need not preserve exact synchrony. Its constants are independent of the number of nodes, provided the critical projections, transverse estimates, history traces, model-fitting map, and simple-root bounds are uniform. This is a local selected-root theorem, not a claim about arbitrary graph sequences.

Third, a shared-resource heterogeneous-curvature class realizes the mechanism on finite directed Markov networks with a common Dobrushin gap and without a nontrivial synchrony quotient. Fixed delay support and a declared row-neutral perturbation are essential. Sparse or closing-gap families and arbitrary changes of the base topology are outside the proved scope.

1.2 Contributions and claim boundary

The theorem-level contributions currently assembled in this draft are:

(i) a dimension- and topology-uniform transverse decay estimate and causal complete-line Green bound for every finite leaky Dobrushin network with a nonnegative row-stochastic current layer and fixed nonnegative half-row-mass delay layers in the declared voltage strip, without common delay-layer left balance;

(ii) an exact upper-triangular canonical Lin factorization, bounded invertible codomain row reduction to the scalar/transverse direct sum, and corrected full cokernel, together with a general normalized one-gap perturbation theorem having an explicit root radius and quadratic remainder;

(iii) a synchrony-quotient-free shared-resource network whose selected complete-history root has a nonzero, dimension-uniform topology-resolvent coefficient;

(iv) validated scalar leaky-recovery invariant objects: a stable quiet equilibrium, two periodic RFDE branches, exactly one nontranslation unstable multiplier for the center inner orbit, and a directed nonsingular frequency–amplitude response, together with a continuous RFDE adjoint and a correlated physical-time base-orbit uu stable-output bound, a direct common-row terminal certificate for the selected phase-fixed inner linear map with $\|AP_s\|_{\Sigma_0 \to Y} \leq 0.009896427481610001 < 0.1$, yielding $K_s = 1$ at the registered rate $\rho_s = 0.1$, and an event-aware signed-density proof that the phase-fixed outer linear return contracts arbitrary reduced C^0 section histories with norm at most 0.550516, followed by a source-bound nonlinear local return and attracting tube with section radius $r_6 = 10^{-335}$, full complete-history tube radius $R_6 = 10^{-3}$, and return constant $\Lambda_6 \le 0.561839$;

(v) a general eventual-smoothing theorem for selected complete-history RFDE events, together with a model-specific common near-two-period event tube on the explicit arbitrary-C^0-history scale $\lambda_0 = 1.2 \times 10^{-8}$, obtained from an orbit-aware weighted energy estimate, the direct derivative identity $DQ_Y(Y_*) = A^2$, transferred two-step hyperbolic rates, and a finite- sampling obstruction showing why nodal refinement alone cannot certify the required Banach-space Hessian;

(vi) a wide-parameter physical-pulse theorem with a unique transverse Route–C event, a fourth-order event-time graph, and a continuous complete-history tube on the whole parameter interval, together with a direct event-aligned first-variation theorem retaining the moving-time contribution and a fixed-center, physical-right-gauge-corrected normalized action that has one strict sign throughout the pulse interval, together with source-bound endpoint functional signs, a full-interval stable-coordinate ball, and a strengthened conditional stable-gap slope gate in the same section, history norm, and Grushin normalization;

(vii) an event-aligned stable-sheet theorem and a block-triangular frequency–amplitude–threshold-offset inverse, conditional on explicit numerical graph and crossing inequalities; its interpretation as a safety coordinate additionally requires the stated routing inequalities.

Item (vii) is not promoted to a model theorem in the present research draft. The discrete stable-power input required by a future inner graph transform is now proved for the selected near-one-period phase-fixed linear map, with $K_s = 1$ and $\rho_s = 0.1$. The selected two-period map is now defined and C^2 on a microscopic nonzero scaled ball, and its center derivative, fixed splitting, and two-step rates are proved. The remaining inner certificates are its enlargement to the preferred graph domain, return-domain containment there, all six uniform split-ball Hessian blocks, and a quantitative stable graph whose domain contains the certified radius-47/5000 stable-coordinate ball, together with its derivative and endpoint-height bounds. The stable-coordinate ball itself and the conditional strict-slope implication are proved; graph-adjusted endpoint signs and the selected crossing are not. Entry of the $J = 0.32$ history into the microscopic outer return domain, outer-side attachment, and two-sided basin routing also remain open. The proved Route–C event is a common history-space section, not an ordinal claim about the third post-release crossing. A global outer Floquet count is useful but is not logically required if local attraction is proved directly. Conversely, none of the network root results identifies the selected canard root with the physical pulse onset.

Frequency–amplitude reduction and control of biological oscillators motivate the proposed final coordinate system (Wilson and Moehlis, 2016; Kotani et al., 2020; Qin et al., 2021; Zhong et al., 2023). Its third target coordinate is the signed offset from a selected history-space stable-sheet threshold, not an independently imposed detector variable. Only the still-open uniform routing theorem would promote that offset to a biological safety margin.

1.3 Organization

Section 2 introduces the scalar RFDE, finite network class, and theorem package. Subsequent sections will supply the complete-line inverse, structural root transfer, validated scalar objects, physical-pulse separator, and network robustness. Directed arithmetic, source manifests, and hostile mutation tests belong to the reproducibility appendices rather than to the mechanism-level argument.

2 Model, network class, and main results

2.1 Scalar leaky-recovery RFDE

Let $r = 5\sqrt{5}$, $(\tau_0, \tau_1) = (4\sqrt{5}, 5\sqrt{5})$, and $H(v) = (v-1)^3$. The autonomous scalar model is

$$\dot{v} = v - \frac{v^3}{3} - w + \varepsilon\kappa_1 \left\{ \frac{v(t-\tau_0) + v(t-\tau_1)}{2} - v(t) \right\} + \varepsilon\kappa_3 \left\{ \frac{H(v(t-\tau_0)) + H(v(t-\tau_1))}{2} - H(v(t)) \right\}, \tag{1}$$

$$\dot{w} = \varepsilon(v - a - w). \tag{2}$$

The validated center is

$$\varepsilon = \frac{1}{5}, \qquad a = \frac{1}{4}, \qquad \kappa_1 = \frac{1}{250}, \qquad \kappa_3 = \frac{1}{200}. \tag{3}$$

A physical pulse of amplitude J adds $J\mathbf{1}_{[0,1)}(t)$ to equation (1); after release the vector field is exactly autonomous again.

The full phase space is $X = C([-r, 0], \mathbb{R}^2)$. Because the recovery equation has no delayed recovery slot, the positive-time semiflow factors through

$$Y = C([-r, 0], \mathbb{R}) \times \mathbb{R}, \qquad \Pi_Y(\phi_v, \phi_w) = (\phi_v, \phi_w(0)). \tag{4}$$

This reduction preserves every nonzero Floquet multiplier and pulls stable sets back exactly; an obsolete recovery history is not an additional crossing coordinate.

2.2 Finite row-mass Dobrushin networks

For a finite network let $Q, B_0, B_1 \in \mathbb{R}^{N\times N}$ be nonnegative matrices and let π be any stationary probability vector satisfying

$$Q\mathbf{1} = \mathbf{1}, \quad \pi^T Q = \pi^T, \quad \pi^T\mathbf{1} = 1, \quad \tau(Q) \le \frac{1}{2}, \qquad B_j\mathbf{1} = \frac{1}{2}\mathbf{1}. \tag{5}$$

Here $\tau(Q)$ is the Dobrushin coefficient. Zero entries of π are allowed, and no lower bound on $\pi_{\min}$ enters the estimates. The row-mass identities preserve the collective line $\operatorname{span}\{\mathbf{1}\}$, but $\operatorname{Ker} \pi^T$ need not be invariant under the delayed layers: no identity $\pi^T B_j = \frac{1}{2}\pi^T$ is assumed. Collective/quotient coordinates therefore produce an upper-triangular operator. The former balanced direct sum is the special case

$$\delta_B = 0, \qquad \delta_B := \delta_0 + \delta_1, \qquad \delta_j := \frac{1}{2}\left\|\pi^T B_j - \tfrac{1}{2}\pi^T\right\|_1, \qquad 0 \le \delta_B \le 1.$$

The finite leaky network to which the estimates below apply is, with $f(s) = s - s^3/3$ and both f and H acting componentwise,

$$\dot{v}(t) = f(v(t)) - w(t) + 3(Q - \mathrm{I})v(t) + \varepsilon\kappa_1\{B_0 v(t-\tau_0) + B_1 v(t-\tau_1) - v(t)\} + \varepsilon\kappa_3\{B_0 H(v(t-\tau_0)) + B_1 H(v(t-\tau_1)) - H(v(t))\}, \tag{6}$$

$$\dot{w}(t) = \varepsilon\{v(t) - a\mathbf{1} - w(t)\} + 2(Q - \mathrm{I})w(t). \tag{7}$$

Here $v, w \in \mathbb{R}^N$, and all scalar constants are exactly those in equation (3). A collective physical pulse adds $J\mathbf{1}_{[0,1)}(t)\mathbf{1}$ to equation (6). The row-mass identities imply that $v = V\mathbf{1}$, $w = W\mathbf{1}$ is invariant and that its restriction is exactly equations (1) and (2), including the two half-weighted delayed terms. Thus the constants in the transverse theorem below refer to this RFDE, not to an unspecified network realization.

Use the transverse history diameter norm

$$\|\phi\|_{1/10} = \sup_{-r\le\theta\le 0} e^{\theta/10} \max\{\operatorname{osc}\phi_v(\theta), 3\operatorname{osc}\phi_w(\theta)\}. \tag{8}$$

Theorem 2.1 (Complete-line transverse inverse)**.** *Along every complete synchronous leaky trajectory satisfying* $\sup_t |V(t) - 1| \le 5/2$*, every network satisfying equation* (5) *has*

$$\|U_{\perp,N}(t,s)\|_{1/10} \le e^{-(t-s)/10}, \qquad \|G_{\perp,N}\| \le 10. \tag{9}$$

The constants are independent of the finite network size and admitted topology.

The proof uses a directed Dobrushin–Halanay estimate and a forward pullback of zero-history forced solutions. It never inverts an RFDE semiflow on the whole history space.

Corollary 2.2 (Upper-triangular canonical root transfer)**.** *Suppose a scalar phase-fixed Lin operator has Fredholm index* -1*, a one-dimensional normalized cokernel, gap* $d(\nu)$*, and a simple root* $d(\nu_c) = 0$*. If all network endpoint, phase, parameter, and gap rows are collective, then*

$$\mathcal{L}_N = \begin{pmatrix} \mathcal{L}_\| & C_N \\ 0 & \mathcal{L}_{\perp,N} \end{pmatrix}, \qquad T_N\mathcal{L}_N = \mathcal{L}_\| \oplus \mathcal{L}_{\perp,N}, \tag{10}$$

where

$$T_N(y_\|, y_\perp) = \big(y_\| - C_N G_{\perp,N} y_\perp, y_\perp\big), \qquad \|C_N G_{\perp,N}\| \le \frac{391}{2000}\delta_B.$$

If ψ *is the normalized scalar cokernel, the full normalized cokernel is*

$$\Psi_N(y_\|, y_\perp) = \psi\big(y_\| - C_N G_{\perp,N} y_\perp\big). \tag{11}$$

For a collective parameter row $(g_\|, 0)$*, one has* $\Psi_N(g_\|, 0) = \psi(g_\|)$*. Hence* $d_N(\nu) = d(\nu)$*, and every admitted finite network has the same Fredholm index, root, slope, and orientation as the scalar problem.*

For the leaky model, theorem 2.1 verifies the network block of this corollary. The result is conditional on the scalar phase-fixed complete-history root and the declared collective preparation. The scalar leaky root itself, an invariant asynchronous voltage strip, and physical pulse onset remain separate open problems.

2.3 Uniform structural transfer

Let $\widetilde{d}_N(\nu, \mathcal{R})$ be a normalized selected gap on a common parameter cylinder. The residual norm is an operator norm controlling the instantaneous block, fixed-support delay measure, nonlinear jets, and trace preparation; it is not an unscaled Frobenius norm.

Theorem 2.3 (Dimension-uniform one-gap theorem)**.** *Assume the graph–trace or augmented-Lin range problem is uniformly solvable, the model-fitting error is at most* $C_{\mathrm{fit}}\|\mathcal{R}\|$*, and* $\widetilde{d}_N$ *is jointly* C^2*. Suppose, uniformly in* N*,*

$$\widetilde{d}_N(\nu_{0,N}, 0) = 0, \qquad |a_N| := \left|\partial_\nu \widetilde{d}_N(\nu_{0,N}, 0)\right| \ge m_* > 0, \qquad \left\|D_\mathcal{R}\widetilde{d}_N\right\| \le B_*, \qquad \left\|D^2\widetilde{d}_N\right\| \le M_*. \tag{12}$$

Here $B_*, M_*, C_{\mathrm{fit}} \geq 0$. If $r_\nu, r_{\mathcal{R}}, \eta_* > 0$ are respectively the common cylinder radii and the range-solve radius, set

$$\rho_* = \min\left\{ r_{\mathcal{R}}, \frac{m_* r_\nu}{2B_*}, \frac{m_*}{2M_*(1+2B_*/m_*)}, \frac{\eta_*}{C_{\mathrm{fit}}} \right\}. \tag{13}$$

Every quotient in equation (13) whose nonnegative denominator is zero is, by convention, $+\infty$; hence an identically absent error term imposes no artificial radius restriction. For every $r = \|\mathcal{R}_N\| \leq \rho_*$, there is exactly one root in

$$|\nu - \nu_{0,N}| \leq \frac{2B_*}{m_*} r, \tag{14}$$

and, with $\ell_N = D_{\mathcal{R}} \widetilde{d}_N(\nu_{0,N}, 0)$,

$$\left| \nu_{c,N}(\mathcal{R}_N) - \nu_{0,N} + \frac{\ell_N[\mathcal{R}_N]}{a_N} \right| \leq \frac{M_*}{2m_*} \left(1 + \frac{2B_*}{m_*}\right)^2 r^2. \tag{15}$$

All constants are independent of the admitted finite network size.

The theorem concerns a selected complete-history root. A zero-fiber implication is additionally required to call it a geometric canard, and a basin theorem is required to call anything a physical pulse threshold.

2.4 A synchrony-quotient-free instance

We next state a separate network result whose node fields are heterogeneous and whose slow recovery variable is shared. For each $N \geq 2$, let $P_N \geq 0$ be row stochastic and let π_N be a strictly positive stationary probability vector. For fixed $D, \gamma > 0$, assume

$$P_N \mathbf{1} = \mathbf{1}, \qquad \pi_N^T P_N = \pi_N^T, \qquad \pi_N^T \mathbf{1} = 1, \qquad \tau(P_N) \leq 1 - \gamma. \tag{16}$$

Set

$$P_{c,N} = \mathbf{1}\pi_N^T, \qquad P_{\perp,N} = \mathrm{I} - P_{c,N}, \qquad E_N = \operatorname{Ker} \pi_N^T, \qquad A_N = D(P_N - \mathrm{I})|_{E_N}, \tag{17}$$

and equip $\mathbb{R}^N$ with

$$\|x\|_N = \left|\pi_N^T x\right| + \operatorname{osc}(x). \tag{18}$$

Let the curvature vectors satisfy, with constants independent of N,

$$0 < c_- \leq c_{i,N} \leq c_+, \qquad \pi_N^T c_N = \alpha > 0. \tag{19}$$

Fix $\beta > 0$, $K \neq 0$, and finitely many delay locations $0 \leq \theta_0 < \cdots < \theta_m \leq \Theta_*$. For

$$B_{k,N}(\zeta) = B_{k,N} + \zeta R_{k,N}, \qquad C_N(\zeta) = \sum_{k=0}^{m} B_{k,N}(\zeta), \tag{20}$$

assume the fixed-support operator bound and the atomwise full-row neutrality

$$\sup_N \sum_{k=0}^{m} \left(\|B_{k,N}\|_{\mathcal{L}(\mathbb{R}^N, \|\cdot\|_N)} + \|R_{k,N}\|_{\mathcal{L}(\mathbb{R}^N, \|\cdot\|_N)} \right) < \infty, \qquad \pi_N^T R_{k,N} = 0. \tag{21}$$

The latter is a row identity, not merely $\pi_N^T R_{k,N} \mathbf{1} = 0$.

For $\varepsilon = \delta^2$, the RFDE is

$$\dot{v}(t) = \frac{2}{3}\mathbf{1} - w(t)\mathbf{1} - c_N \circ (v(t) - \mathbf{1})^{\circ 2} - \frac{\beta}{3}(v(t) - \mathbf{1})^{\circ 3} + D(P_N - \mathrm{I})v(t) \\ + \delta^2 K \left\{ C_N(\zeta)v(t) - \sum_{k=0}^{m} B_{k,N}(\zeta) v\left(t - \frac{\theta_k}{\delta}\right) \right\}, \tag{22}$$

$$\dot{w}(t) = \delta^2\{\pi_N^T v(t) - a\}, \tag{23}$$

where $v \in \mathbb{R}^N$, $w \in \mathbb{R}$, and the powers and products are componentwise. Its collective fold is $(v, w, a) = (\mathbf{1}, 2/3, 1)$. Write

$$a = 1 + \delta^2\nu, \qquad \mu = a - 1 = \delta^2\nu, \qquad \dot{M}_{1,N} = \sum_{k=0}^{m} \theta_k R_{k,N}, \tag{24}$$

and define

$$g_N = A_N^{-1} P_{\perp,N} c_N,$$

$$\kappa_N = \frac{\beta}{3} + 2\pi_N^T \operatorname{diag}(c_N) g_N, \qquad \nu_{0,N} = -\frac{3\kappa_N}{8\alpha^3}, \tag{25}$$

$$r_N = \frac{K}{2\alpha}\pi_N^T \operatorname{diag}(c_N) A_N^{-1} P_{\perp,N} \dot{M}_{1,N}\mathbf{1}. \tag{26}$$

For completeness, we also fix the selection entering the theorem. In the fold chart

$$s = \delta t, \qquad w = \frac{2}{3} - \delta^2 Y, \qquad v = \mathbf{1} + \delta\mathbf{1}X + \delta^2\{g_N X^2 + h\}, \qquad h \in E_N,$$

the singular critical field is $q_0(X, Y) = (Y - \alpha X^2, -X)$. A uniformly admissible preparation rule $\mathcal{P} = (\mathcal{P}_N)$ means that the same frozen graph cutoffs, planar joining cutoff, degree-three normal extension, phase buffer, and invariant-tail levels are used for every N, with common derivative bounds; the preparation is independent of (ν, ζ) and agrees with equations (22) and (23) on the retained continuous depth-two q_0-flow hull of half-length $S_\delta = \sqrt{2p\log(1/\delta)}$, for an exponent $p > 4$ fixed in the theorem below. It selects attracting and repelling phase-fixed planar traces $z^a_{N,\mathcal{P}_N}$ and $z^r_{N,\mathcal{P}_N}$, with phase $X(0) = 0$. Their normalized central gap is

$$\mathscr{D}_{N,\mathcal{P}_N}(\delta, \nu, \zeta) = \frac{2}{\alpha e}\left\{\mathscr{H}_\alpha(z^a_{N,\mathcal{P}_N}(0)) - \mathscr{H}_\alpha(z^r_{N,\mathcal{P}_N}(0))\right\}, \tag{27}$$

$$\mathscr{H}_\alpha(X, Y) = \frac{1}{2}e^{-2\alpha Y}\left(\alpha Y - \alpha^2 X^2 + \frac{1}{2}\right). \tag{28}$$

On the retained graph, a zero of $\mathscr{D}_{N,\mathcal{P}_N}$ is equivalent to equality of the two selected complete RFDE histories.

Theorem 2.4 (Shared-resource network response)**.** *Assume equations* (16)*,* (19) *and* (21) *with all displayed bounds common in N, and assume*

$$\inf_{N \geq 2} |r_N| > 0. \tag{29}$$

Fix $p > 4$ sufficiently large, a compact interval I_ν containing every $\nu_{0,N}$ in its interior with one common positive boundary distance, and a class of uniformly admissible preparation rules with one *set of bounds. Then there exist $\delta_0, \zeta_0, c_0, C > 0$, independent of N and of the rule in that class, such that for every $N \geq 2$, every $0 < \delta < \delta_0$, every $|\zeta| < \zeta_0$, and every admitted $\mathcal{P}$, the gap $\mathscr{D}_{N,\mathcal{P}_N}/\delta$ has exactly one root in the local window $\{\nu : |\nu - \nu_{0,N}| < c_0\} \subset I_\nu$,*

$$\nu_{c,N,\mathcal{P}_N}(\delta, \zeta) \in I_\nu, \qquad |\nu_{c,N,\mathcal{P}_N}(\delta, \zeta) - \nu_{0,N}| < c_0. \tag{30}$$

No assertion is made here about additional zeros elsewhere in I_ν. The selected attracting and repelling complete histories agree at this root. With $\mu_{c,N,\mathcal{P}_N} = \delta^2 \nu_{c,N,\mathcal{P}_N}$,

$$\left|\mu_{c,N,\mathcal{P}_N}(\delta, \zeta) - \mu_{c,N,\mathcal{P}_N}(\delta, 0) - \mathscr{C}_N \delta^3 \zeta\right| \leq C\{\delta^4 |\zeta| + \delta^3 \zeta^2\}, \tag{31}$$

$$\mathscr{C}_N = -\frac{r_N}{\alpha} = -\frac{K}{2\alpha^2} \pi_N^T \operatorname{diag}(c_N) A_N^{-1} P_{\perp,N} \dot{M}_{1,N} \mathbf{1}. \tag{32}$$

In particular, $\inf_N |\mathscr{C}_N| > 0$. The exact finite-$\delta$ root may depend on $\mathcal{P}_N$; its coefficient and the displayed remainder bound do not within the declared preparation class.

The nondegeneracy class is nonempty without a hidden synchrony quotient. Indeed, take $\pi_N = N^{-1}\mathbf{1}$, $P_N = P_{c,N}$, and

$$s_{i,N} = \frac{\sqrt{3}(2i - N - 1)}{\sqrt{(N-1)(N+1)}}, \qquad c_N = \mathbf{1} + \sigma s_N, \qquad 0 < \sigma < 1/\sqrt{3}.$$

These choices satisfy

$$\pi_N^T s_N = 0, \qquad \pi_N^T s_N^{\circ 2} = 1, \qquad \|s_N\|_\infty < \sqrt{3}, \qquad \alpha = 1.$$

For two delays $\theta_0 < \theta_1$, set

$$B_{0,N} = B_{1,N} = bP_{c,N}, \qquad R_{0,N} = s_N \pi_N^T, \qquad R_{1,N} = -s_N \pi_N^T, \qquad b > 0.$$

The layers remain entrywise positive for $|\zeta| < b/\sqrt{3}$, all entries of c_N are distinct, and

$$\mathscr{C}_N = \frac{K\sigma(\theta_0 - \theta_1)}{2D} \neq 0 \tag{33}$$

for every $N \geq 2$. The distinct curvatures destroy every nontrivial synchrony polydiagonal.

This instance proves that the general root mechanism is not merely an arbitrary-size replication of a two-module quotient. It does not cover moving delay atoms, arbitrary perturbations of the base topology, or graph families whose mixing gap closes, and it does not identify the preparation-indexed root with an independently selected physical outer canard.

2.5 Eventually smooth selected returns

The next result is independent of the preceding network reduction. It isolates the regularity needed to turn a selected event into a moving *complete-history* return. This distinction matters because a fixed-time RFDE solution map may be smooth in its initial history while its joint dependence on time and an arbitrary continuous history is not yet smooth: before the translation part has cleared the initial interval, changing time moves an evaluation point in a merely continuous function.

Let

$$\mathcal{X}_{\tau_*} = C([-\tau_*, 0], \mathbb{R}^d), \qquad \dot{x}(t) = \mathcal{F}(x_t), \qquad x_t(\theta) = x(t + \theta), \tag{34}$$

where $d < \infty$, $\tau_* > 0$, and $\mathcal{F} \colon U \subset \mathcal{X}_{\tau_*} \to \mathbb{R}^d$ is C^r. This includes any finite network with finitely many constant delays bounded by τ_*. Write $\Phi_t(\phi) = x_t(\phi)$.

Theorem 2.5 (Eventually smooth selected-event return)**.** *Let $1 \leq k \leq r$, let $\mathcal{M}$ be a Banach space, $D \subset \mathcal{M}$ open, and let $\iota\colon D \to U \subset \mathcal{X}_{\tau_*}$ be C^k. Suppose all solutions from $\iota(D)$ exist in one common U-tube through T_+. Let $V \subset \mathcal{X}_{\tau_*}$ be open, let $g\colon V \to \mathbb{R}$ be C^k, and assume $\Phi_t(\iota(u)) \in V$ for $(t,u) \in I \times D$. Put*

$$\mathcal{H}(t,u) = g(\Phi_t(\iota(u))).$$

Suppose a common interval $I = [T_-, T_+]$ satisfies

$$T_- > k\tau_*, \qquad \sup_{u\in D} \mathcal{H}(T_-,u) \leq -\delta_-, \qquad \inf_{u\in D} \mathcal{H}(T_+,u) \geq \delta_+, \qquad \partial_t \mathcal{H} \geq a_* > 0 \tag{35}$$

on $I \times D$, for $\delta_-, \delta_+, a_* > 0$. Then each $u \in D$ has exactly one selected event $T(u) \in (T_-, T_+)$, and the event time and ambient event-hit map

$$T\colon D \to \mathbb{R}, \qquad \mathcal{E}(u) = \Phi_{T(u)}(\iota(u))\colon D \to \mathcal{X}_{\tau_*}$$

are C^k. The threshold $k\tau_*$ is a uniform sufficient condition; no necessity or optimality is asserted. It is independent of the number of nodes in any finite network, although concrete tube and derivative constants need not be.

If ι is a C^k chart of a local section Σ, and $\mathcal{E}(D_0) \subset \iota(D)$ for an open $D_0 \subset D$, then

$$\mathcal{Q} = \iota^{-1} \circ \mathcal{E}\big|_{D_0}\colon D_0 \to D \tag{36}$$

is the induced C^k selected section-return map.

Proof. On the common tube, repeated Fréchet differentiation of the Volterra equation gives, for every fixed time, the triangular initial-data variational hierarchy through order k. We record the additional joint smoothing step because it is essential here. Inductively in j, the mixed jets

$$\partial_t^a D_\phi^b x(t;\phi), \qquad a + b \leq j,$$

exist after $(j-1)\tau_*$ and depend continuously on (t,ϕ) in the corresponding multilinear-operator norms. For $j = 1$ this follows from the integral equation and the first variational equation. If it holds through order j, differentiating $x' = \mathcal{F}(x_t)$ and the triangular variational equations once more expresses every order-$j+1$ jet as a finite sum of continuous derivatives of $\mathcal{F}$ applied to lower-order history jets. After one further maximum delay, every argument segment lies in the already regularized interval. The linear inhomogeneous variational equations and Gronwall's inequality give continuity in the multilinear- operator norms, completing the induction.

The earliest physical time in $\Phi_t(\phi)$ is $t - \tau_*$. Hence, for every total order $j \leq k$, the preceding jets define continuous $\mathcal{X}_{\tau_*}$-valued derivatives once $t > j\tau_*$. Therefore $(t,\phi) \mapsto \Phi_t(\phi)$ is jointly C^k for $t > k\tau_*$, and $\mathcal{H}$ is jointly C^k on $I \times D$. The endpoint signs and positive speed give existence and uniqueness in the declared window; the Banach implicit-function theorem gives $T \in C^k$, and composition gives $\mathcal{E} \in C^k$. The last assertion follows by composing with the section chart inverse. See also the fixed-time RFDE and Poincaré-map background in Diekmann et al. (1995, Chs. VII.4 and XIV.3). □

No exclusion of earlier hits is used in theorem 2.5: it is needed only to call the branch a first return or to assign it an exact ordinal. This separation also avoids an unnecessary algebraic restriction on a stable-manifold construction.

Lemma 2.6 (Direct selected return identifies the stable-set germ)**.** *Let Φ be a continuous semiflow, Γ a periodic orbit through a transverse local section Σ at p. Choose a local section patch $N \subset \Sigma$ with the phase-isolation property*

$$x_n \in N, \qquad \operatorname{dist}(x_n, \Gamma) \to 0 \quad \Longrightarrow \quad x_n \to p. \tag{37}$$

Let $\mathcal{Q}(x) = \Phi_{\Theta(x)}(x)$ be a selected local return of the same semiflow with

$$\mathcal{Q}\colon N \to N, \qquad \mathcal{Q}(p) = p, \qquad \Theta(p) = mP, \qquad 0 < \Theta_- \le \Theta(x) \le \Theta_+ < \infty.$$

Suppose every intervening arc between successive iterates stays in one common flow tube. Restrict both stable sets to points whose relevant iterates or flow arcs stay in these declared local domains, and assume that local flow-stable points in N have the selected hits recurrently defined in N. Then, as germs at p,

$$\operatorname{germ}_p W^s_N(\mathcal{Q}) = \operatorname{germ}_p\big(N \cap W^s_{\text{tube}}(\Gamma)\big). \tag{38}$$

Consequently, a direct near-mP selected return can construct the intrinsic periodic-orbit stable-sheet germ; an identity $\mathcal{Q} = P^m$ and a first-return label are sufficient conveniences, not necessary hypotheses. If $\mathcal{Q}$ is C^2, p has the required hyperbolic splitting, and a local graph theorem applies, its stable graph represents this same intrinsic germ.

Proof. Put $t_n = \sum_{j=0}^{n-1} \Theta(\mathcal{Q}^j x)$. If $\mathcal{Q}^n(x) \to p$, then $t_n \to \infty$ by the positive lower bound; joint continuity of Φ_t, uniformly for $0 \le t \le \Theta_+$, sends every intervening late-time arc to the corresponding arc of Γ. Conversely, convergence of the flow orbit to Γ, together with recurrent selected hits in N, gives $\operatorname{dist}(\mathcal{Q}^n x, \Gamma) \to 0$. The phase-isolation property equation (37) then gives $\mathcal{Q}^n x \to p$. □

The theorem deliberately excludes state-dependent or neutral delays, infinite delays, infinite networks, and infinite-dimensional node states. It proves a reusable return-map mechanism, not a return tube or event window for the concrete leaky model.

2.6 Conditional physical-pulse and control theorem

Let $\xi = (a, \kappa_3)$ range in a closed rectangle $\Xi \subset \mathbb{R}^2$ about $\xi_0 = (1/4, 1/200)$, with $\varepsilon, \kappa_1, \tau_0, \tau_1$ fixed as in equation (3). Write

$$\bar v_q(a) = (3a)^{1/3}, \qquad E_q(\xi)(\theta) \equiv \big(\bar v_q(a), \bar v_q(a) - a\big), \quad -r \le \theta \le 0, \tag{39}$$

for the quiet equilibrium history in X, and let Φ^ξ_t be the autonomous post-release semiflow. Let $\Gamma_i(\xi)$ and $\Gamma_o(\xi)$ be the inner and outer scalar periodic orbits, identified with their compact orbits of histories in X, and set

$$F(\xi) = T_o(\xi)^{-1}, \qquad A(\xi) = \max_t V_o(t;\xi) - \min_t V_o(t;\xi). \tag{40}$$

For a physical one-unit pulse of amplitude J, applied from $E_q(\xi)$, denote the history at release by $R_\xi(J) \in X$. On a phase section $\Sigma_\xi \subset Y$ through $\Gamma_i(\xi)$, let $B_\xi(J)$ be the event-aligned entrance of the post-release orbit. In section coordinates $(u, z) \in E^u_\xi \oplus E^s_\xi$, write the local stable sheet as

$$W^s_{\text{loc}}(\Gamma_i(\xi)) \cap \Sigma_\xi = \{(u, z) : u = \psi_\xi(z)\}, \qquad H(\xi, J) = u_\xi(J) - \psi_\xi(z_\xi(J)), \tag{41}$$

where $(u_\xi(J), z_\xi(J))$ are the coordinates of $B_\xi(J)$.

Theorem 2.7 (Pulse threshold and safety coordinates: completion theorem)**.** *Assume the following objects and estimates hold with common domains, radii, and strict constants for every $\xi \in \Xi$. The equilibrium $E_q(\xi)$ and the phase-fixed branches $\Gamma_i(\xi), \Gamma_o(\xi)$ are C^1 in ξ; the quiet equilibrium and outer cycle possess quantitative attraction tubes; and the outer voltage extrema are simple. The inner section return has a one-dimensional unstable bundle and a stable complement, both C^1 in ξ, and $\psi_\xi(z)$ in equation (41) is jointly C^1 in (ξ, z) on one common graph domain.*

Assume also that $(\xi, J) \mapsto R_\xi(J)$ is C^1 on $\Xi \times I_J$, where $I_J = [J_-, J_+]$, and that $B_\xi(J)$ is one jointly C^1, event-aligned entrance branch on this whole product: the section event does not switch branches, its speed is bounded away from zero, and every $B_\xi(J)$ lies in the common domain of the signed-exit construction, while every stable coordinate $z_\xi(J)$ lies in the common domain of ψ_ξ. Suppose, after fixing the orientation, that there are $\eta_H, m_J > 0$ such that

$$H(\xi, J_-) \geq \eta_H, \qquad H(\xi, J_+) \leq -\eta_H, \qquad \partial_J H(\xi, J) \leq -m_J < 0, \tag{42}$$

for all $\xi \in \Xi$ and $J \in I_J$. In particular, H is a jointly C^1 function on this full product, rather than a collection of pointwise gaps. Finally, assume the signed-exit dynamics sends the $H > 0$ side into a complete-history exit slab attached to the quiet attraction tube and the $H < 0$ side into a complete-history exit slab attached to the outer attraction tube, uniformly in ξ, with no other local exit; histories on $H = 0$ lie on $W^s_{\mathrm{loc}}(\Gamma_i(\xi))$.

Then there is a unique C^1 function $J_c : \Xi \to (J_-, J_+)$ satisfying $H(\xi, J_c(\xi)) = 0$, with

$$D_\xi J_c(\xi) = -\frac{D_\xi H(\xi, J_c(\xi))}{\partial_J H(\xi, J_c(\xi))}. \tag{43}$$

For every $(\xi, J) \in \Xi \times I_J$, the routing assertions mean the following asymptotic statements in the complete history space:

$$\begin{aligned} J < J_c(\xi) &\implies \left\| \Phi_t^\xi R_\xi(J) - E_q(\xi) \right\|_X \longrightarrow 0, \\ J = J_c(\xi) &\implies \mathrm{dist}_X(\Phi_t^\xi R_\xi(J), \Gamma_i(\xi)) \longrightarrow 0, \qquad (t \to \infty). \\ J > J_c(\xi) &\implies \mathrm{dist}_X(\Phi_t^\xi R_\xi(J), \Gamma_o(\xi)) \longrightarrow 0 \end{aligned} \tag{44}$$

Define the signed intrinsic safety margin $S(\xi, J) = J - J_c(\xi)$. If $\xi_* \in \operatorname{int} \Xi$ satisfies $\det D_\xi(F, A)(\xi_*) \neq 0$, then, for every $J_* \in \operatorname{int} I_J$,

$$\mathcal{Q}(\xi, J) = (F(\xi), A(\xi), S(\xi, J)) \tag{45}$$

is a local C^1 diffeomorphism at (ξ_*, J_*), and

$$\det D\mathcal{Q}(\xi_*, J_*) = \det D_\xi(F, A)(\xi_*). \tag{46}$$

The implication in theorem 2.7 is proved by a forward Lyapunov–Perron stable graph, graph-adjusted stable-gap endpoint signs plus strict gap monotonicity, two-sided routing, and the block-triangular inverse theorem. Interval Newton may sharpen the resulting scalar root enclosure but is not needed for its existence or uniqueness. The model-specific graph and routing premises are current release gates. What is already proved at the center is the exact reduced-history factorization, strict quiet capture at $J = 0.30$, exactly one nontranslation unstable inner multiplier, an explicit stable spectral gap, a continuous Route–C adjoint history measure with nonzero normalization, a physical-time base-orbit correlated second-return bound $C^{uu}_{s,\mathrm{base}} < 7.905650 < 12$, directed outer (F, A) nonsingularity, a rigorous fixed-time fourth-order pulse-parameter model with fifth-order remainder on $J \in [0.30105, 0.30120]$, and an ambient complete-history proximity bound at

$J = 0.32$, not a section/domain attachment. On that pulse interval a separate validated event theorem also gives a unique positive Route–C crossing in a common bracket, a fourth-order event-time graph, and a continuous common-event history tube; see theorem 4.12. Its event-aligned complete-history first derivative, including the moving-time term, is also enclosed on the full interval in theorem 4.13. After correcting the complex right eigenvector gauge, theorem 4.16 proves that the fixed normalized Route–C functional has a strictly negative action on this derivative throughout the interval. In the same continuous-history norm, Route–C section, physical Grushin normalization, and centered coordinate $\kappa(J) = K(J) - X_*$, source-bound endpoint calculations prove $f_{\text{phys}}(\kappa(J_-)) > 0 > f_{\text{phys}}(\kappa(J_+))$. The direct complete-history norm of the same q_{phys} gives $\|P_s D_J \kappa\|_Y \leq 5.979324$, and a two-ended cone proves $P_s \kappa(I_J) \subset \overline{B}_Y(0, 47/5000) \cap \ker f_{\text{phys}}$. Conditional on a quantitative graph whose stable domain contains this ball and $\sup \|D\psi\| \leq 16$, it gives $H'(J) \in [-354.415700, -149.584300] \subset (-\infty, 0)$ on the full pulse interval. The general return theorem theorem 2.5 identifies the missing C^2 adapter: the one-period center does not clear two maximum delays, whereas the near-two-period center does so with margin greater than 14.010740. Theorem 4.5 now proves a common C^2 selected event and complete-history hit on the explicit arbitrary- continuous-history scale $\lambda_0 = 9 \times 10^{-31}$, and the orbit-aware enlargement in theorem 4.6 raises the proved closed scale to 1.2×10^{-8}, while theorem 4.7 proves directly that its center derivative is A^2. For that route, the conditional six-block box in theorem 4.9 would give the sharper physical graph budgets and $H'(J) \in [-260.794147, -243.205853]$, together with strict graph-adjusted endpoint margins. Its signed finite-section pilot lies comfortably inside all six caps, but is not interval evidence. Thus the functional endpoint signs, pulse ball containment, enlarged but still sub-preferred two-period event tube, center derivative, fixed-splitting/rate transfer, and conditional graph/crossing arithmetic are proved. Enlargement of the return to the preferred graph ball, graph-transform domain containment, six directed continuous-history Hessian bounds, stable-set identification, the graph, selected crossing, onset, and routing remain open. Interval Newton is only an optional later sharpening. Neither that observable event and derivative nor the preceding invariant-object facts imply equation (42) or equation (44).

3 The collective–transverse mechanism

This section isolates the mechanism behind theorems 2.1 to 2.3. The point is not that a synchronous orbit can be copied into N nodes; that observation alone gives no uniform inverse and no control of nonsynchronous histories. The useful fact is that the collective RFDE and every transverse network direction admit estimates in norms whose constants do not grow with N.

3.1 Diameter contraction and the delay margin

Write a variational solution along a complete synchronous trajectory as

$$(x, y) = (X\mathbf{1}, Y\mathbf{1}) + (x_\perp, y_\perp), \qquad \pi^T x_\perp = \pi^T y_\perp = 0.$$

For a complex vector z, put $\operatorname{diam} z = \max_{i,k} |z_i - z_k|$. This is a norm on $\operatorname{Ker} \pi^T$. The Dobrushin identities imply

$$\operatorname{diam}(Qz) \leq \tau(Q) \operatorname{diam} z, \qquad \operatorname{diam}(B_j z) \leq \frac{1}{2} \operatorname{diam} z. \tag{47}$$

No eigenbasis, reversibility, or simultaneous diagonalization is used.

Let

$$M(t) = \max\{\operatorname{diam} x_\perp(t), 3 \operatorname{diam} y_\perp(t)\}.$$

On the strip $|V(t)-1| \leq 5/2$, the current and delayed coefficients in the leaky model give

$$\alpha_v \geq 3(1-\tau(Q)) - (1-\varepsilon\kappa_1) - \frac{1}{3} > 0.16746, \tag{48}$$

$$\alpha_w \geq 2(1-\tau(Q)) + \varepsilon - 3\varepsilon \geq 0.6, \tag{49}$$

$$\beta \leq \varepsilon\{\kappa_1 + 3\kappa_3(5/2)^2\} = 0.01955. \tag{50}$$

Consequently

$$D^+M(t) \leq -\alpha M(t) + \beta \sup_{t-r\leq s\leq t} M(s), \qquad \alpha = \min\{\alpha_v, \alpha_w\}. \tag{51}$$

The directed residual

$$\alpha - \frac{1}{10} - \beta e^{r/10} > 0.0076664505356400 \tag{52}$$

is strict. A first-crossing argument for $e^{t/10}M(t)$ proves

$$\|U_{\perp,N}(t,s)\|_{1/10} \leq e^{-(t-s)/10}.$$

This proves equation (9)'s homogeneous part in a norm adapted to node oscillation. It deliberately avoids a dimension-dependent equivalence with a Euclidean product norm.

3.2 A causal complete-line inverse

The inverse in theorem 2.1 is obtained without evolving an RFDE backwards. Given a bounded transverse forcing f, let z^S be the forward solution with zero retained history at time S. The forced version of the same first-crossing argument gives

$$\left\|z_t^S\right\|_{1/10} \leq \int_S^t e^{-(t-u)/10} \|f(u)\|_\perp \, \mathrm{d}u \leq 10 \|f\|_{\perp,\infty}. \tag{53}$$

If $S_1 < S_2 < t$, the two solutions differ homogeneously after S_2; hence

$$\left\|z_t^{S_1} - z_t^{S_2}\right\|_{1/10} \leq 10 e^{-(t-S_2)/10} \|f\|_{\perp,\infty}.$$

The pullback limit $S \to -\infty$ is therefore a unique bounded complete solution and defines $G_{\perp,N}$. This proves $\|G_{\perp,N}\| \leq 10$. The construction is causal and remains valid for nonperiodic complete synchronous trajectories as long as the voltage strip holds.

3.3 Exact transfer and structural transfer

For the canonical synchronized Lin realization, all endpoint, phase, and gap rows are collective. Row mass makes the collective line invariant, but without delay-layer left balance a transverse history can feed the collective equation. Thus the complete-history operator has the exact block form

$$\mathcal{L}_N = \begin{pmatrix} \mathcal{L}_\parallel & C_N \\ 0 & \mathcal{L}_{\perp,N} \end{pmatrix},$$

not in general a direct sum. Since $\mathcal{L}_{\perp,N}^{-1} = G_{\perp,N}$, the bounded codomain row operation

$$T_N(y_\parallel, y_\perp) = \big(y_\parallel - C_N G_{\perp,N} y_\perp, y_\perp\big) \quad \text{satisfies} \quad T_N \mathcal{L}_N = \mathcal{L}_\parallel \oplus \mathcal{L}_{\perp,N}. \tag{54}$$

Consequently the Fredholm index and kernel/cokernel dimensions are inherited from the scalar block. If ψ spans the scalar cokernel, normalized by the collective complement $e_N = (e_\|, 0)$, then the full functional is

$$\Psi_N(y_\|, y_\perp) = \psi\big(y_\| - C_N G_{\perp,N} y_\perp\big). \tag{55}$$

For a collective parameter or inhomogeneity $(g_\|, 0)$, $\Psi_N(g_\|, 0) = \psi(g_\|)$. The normalized root, slope, and orientation are therefore exactly the scalar ones whenever the scalar phase-fixed complete-history root and collective preparation are supplied.

The triangular correction is uniform. With

$$\delta_j = \frac{1}{2}\left\|\pi^T B_j - \tfrac{1}{2}\pi^T\right\|_1, \qquad \delta_B = \delta_0 + \delta_1,$$

oscillation duality and the complete-line Green bound give

$$\|C_N G_{\perp,N}\| \le \frac{391}{2000}\delta_B \le \frac{391}{2000}. \tag{56}$$

No delayed-history residence factor appears here: on complete-line sup norms, a fixed time translation has norm one. This is a statement about complete-history matching. Replacing the history jump by a current-state projection would discard the delayed transverse correction in equation (55).

For structural perturbations, exact splitting is no longer required. We briefly prove the scalar part of theorem 2.3, since it explains the specific radius in equation (13). Put

$$d_N(s, \mathcal{R}) = \widetilde{d}_N(\nu_{0,N} + s, \mathcal{R}).$$

For $r = \|\mathcal{R}\|$, Taylor's formula and equation (12) give, on $|s| \le 2B_* r/m_*$,

$$|\partial_s d_N(s, \mathcal{R}) - a_N| \le M_* \left(|s| + r\right) \le M_* \left(1 + \frac{2B_*}{m_*}\right) r \le \frac{m_*}{2}. \tag{57}$$

Thus the gap is strictly monotone with the orientation of a_N. Moreover, at $s = \pm 2B_* r/m_*$, the linear term dominates the residual, so the endpoint signs are opposite. The first two bounds in equation (13) keep these endpoints inside the common cylinder; the third proves equation (57); and the fourth keeps the graph–trace range solve inside its validated ball. Existence and uniqueness follow.

Writing the root as s_N and expanding at $(0, 0)$ yields

$$0 = a_N s_N + \ell_N[\mathcal{R}] + \mathcal{E}_N, \qquad |\mathcal{E}_N| \le \frac{M_*}{2}(|s_N| + r)^2.$$

The root enclosure then gives equation (15). This calculation also shows why a small residual norm without a uniform simple-root bound is insufficient: the displacement is divided by a_N.

3.4 The shared-resource history graph and selected traces

We now prove the analytic steps used in theorem 2.4. This result has a different role from the preceding triangular transfer: it constructs its own preparation-indexed planar histories for a heterogeneous shared-resource network. In particular, the graph and trace range problems below are not additional unstated hypotheses of the theorem. Here the "invariant-tail levels" in the admissible preparation are the two zero levels $\mathscr{H}_\alpha = 0$ used below. Arbitrary independently chosen outer levels or outer RFDE slow manifolds are not included in this preparation class.

3.4.1 Exact chart and dimension-free model fit

For a complete chart history ϕ, write

$$\mathcal{L}_{N,\zeta}[\phi](s) = \sum_{k=0}^{m}(B_{k,N} + \zeta R_{k,N})\{\phi(s) - \phi(s-\theta_k)\}.$$

Set

$$g_N = A_N^{-1} P_{\perp,N} c_N, \qquad Z_N(X,h) = g_N X^2 + h.$$

Substitution of the fold chart preceding equation (27) into equations (22) and (23) gives the following identities, with no Taylor remainder:

$$\begin{aligned}
X' &= Y - \alpha X^2 \\
&\quad + \delta\left[-2X\pi_N^T \operatorname{diag}(c_N) Z_N - \frac{\beta}{3}X^3 + K\pi_N^T \mathcal{L}_{N,\zeta}[\mathbf{1}X]\right] \\
&\quad + \delta^2\left[-\pi_N^T \operatorname{diag}(c_N) Z_N^{\circ 2} + K\pi_N^T \mathcal{L}_{N,\zeta}[Z_N]\right] - \delta^3\beta X \pi_N^T Z_N^{\circ 2} - \frac{\delta^4\beta}{3}\pi_N^T Z_N^{\circ 3}, && (58)\\
Y' &= -X + \delta\nu, && (59)\\
\delta h' &= A_N h \\
&\quad + \delta\left\{P_{\perp,N}\left[-2X\operatorname{diag}(c_N) Z_N + K\mathcal{L}_{N,\zeta}[\mathbf{1}X]\right] - 2g_N X X'\right\} \\
&\quad + \delta^2 P_{\perp,N}\left[-\operatorname{diag}(c_N) Z_N^{\circ 2} - \beta X^2 Z_N + K\mathcal{L}_{N,\zeta}[Z_N]\right] - \delta^3\beta X P_{\perp,N} Z_N^{\circ 2} - \frac{\delta^4\beta}{3} P_{\perp,N} Z_N^{\circ 3}. && (60)
\end{aligned}$$

The term $-2g_N X X'$ is the derivative of the translation $g_N X^2$. The choice of g_N is forced by $A_N g_N X^2 = P_{\perp,N} c_N X^2$; hence the singular stable graph in these coordinates is $h = 0$, and the reduced singular field is

$$q_0(X,Y) = (Y - \alpha X^2, -X)^T.$$

We record why the constants in this exact chart do not hide a dependence on the number of nodes. The Poisson expansion and Dobrushin contraction give

$$\left\|e^{A_N t}\right\|_{E_N\to E_N} \le e^{-D\gamma t}, \qquad \left\|A_N^{-1}\right\|_{E_N\to E_N} \le (D\gamma)^{-1}. \tag{61}$$

Moreover, in the norm equation (18),

$$\|P_{c,N}\|, \|P_{\perp,N}\| \le 1, \qquad \|x\|_\infty \le \|x\|_N, \qquad \|x_1 \circ \cdots \circ x_j\|_N \le 3\prod_{\ell=1}^{j}\|x_\ell\|_N. \tag{62}$$

Indeed, subtracting $(\pi_N^T x)\mathbf{1}$ does not change oscillation, and a coordinate differs from the π_N-average by at most $\operatorname{osc} x$. The last inequality follows by first estimating each coordinate and then using $\|y\|_N \le 3\|y\|_\infty$. Consequently equations (19) and (21), equation (61), and equation (62) bound every multilinear derivative in equations (58) and (60) by one constant independent of N on a common chart box.

For later use, put

$$\mathcal{X}_N = \mathbb{R}^2 \times E_N, \qquad \mathcal{Y}_N = \mathcal{X}_N^{m+2}$$

with maximum product norms. Current evaluation and the $m+1$ fixed delay evaluations are represented by the fixed atomic measure

$$\mathbf{M}_N = J_{-1}\delta_0 + \sum_{k=0}^{m} J_k \delta_{-\theta_k}, \qquad \|\mathbf{M}_N\|_{\mathrm{TV}} \le m+2, \tag{63}$$

where every insertion J_k has norm one. The matrices depending on ζ remain in the polynomial nonlinearities, so the evaluation measure is parameter independent. This avoids differentiating moving Dirac masses and makes its affine-history anchor derivatives identically zero.

Lemma 3.1 (Uniform graph and trace range)**.** *Fix the common bounds in theorem 2.4 and one bounded class of admissible preparations. After choosing the fixed exponent $p > 4$ sufficiently large, there are $\delta_*, \zeta_*, C, c > 0$ and an integer M, all independent of N and of the preparation in that class, with the following properties.*

For $0 < \delta < \delta_*$, $\nu \in I_\nu$, and $|\zeta| < \zeta_*$, let $S_\delta = \sqrt{2p\log(1/\delta)}$. On a buffered tube about $\gamma_0([-S_\delta, S_\delta])$, the exact system equations (58) and (60) has an injective invariant complete-history graph. Its reduced field has the expansion

$$Q_{N,\delta,\nu,\zeta} = q_0 + \delta q_{1,N} + \delta^2 q_{2,N} + \delta^3 \mathcal{R}_{3,N}, \tag{64}$$

and, for the finite derivative list $a \le 3$, $i \le 1$, $j \le 2$,

$$\left| D_u^a \partial_\nu^i \partial_\zeta^j \mathcal{R}_{3,N}(\gamma_0(s) + \xi) \right| \le C \langle s \rangle^M e^{c|s|} \tag{65}$$

on the nested retained tube. The graph equation and its displayed mixed derivatives have a right inverse bounded uniformly in N.

The prepared tail conditions $\mathscr{H}_\alpha(z^a(-R)) = \mathscr{H}_\alpha(z^r(R)) = 0$, where $R = S_\delta + B$ and $B > 2\Theta_* + 2$ is fixed, together with the phase $X^a(0) = X^r(0) = 0$, select unique one-sided traces. Their retained central pieces are exact histories on the unprepared graph, and their trace–phase– endpoint linearization is onto with a uniformly bounded right inverse. In the tame weighted norms used in the proof,

$$\begin{aligned} \|z^\sigma - \gamma_0\| &\le C\delta, & \|\partial_\nu z^\sigma\| &\le C\delta, \\ \|\partial_\zeta z^\sigma\| + \|\partial_{\zeta\zeta} z^\sigma\| &\le C\delta^2, & \sigma &\in \{a, r\}. \end{aligned} \tag{66}$$

The same $O(\delta^2)$ estimate holds for the mixed derivatives obtained by applying one additional ν-derivative to a ζ-derivative.

Proof. We include the construction because it is the range bridge between the finite-dimensional coefficient calculation and the complete-history root. In the coordinates

$$\chi = -2\alpha X, \qquad d = Y - \alpha X^2 + \frac{1}{2\alpha},$$

prepare q_0 by one common compact cutoff which is the identity on the continuous depth-two q_0-flow hull of the retained tube. Prepare every current and delayed stable slot by the same componentwise smooth saturation followed by $P_{\perp,N}$. Estimate equation (62) shows that all derivatives of these preparations are bounded, without a factor of N, by a polynomial $P(S_\delta)$. The preparation is the physical system on a common stable ball containing the entire retained depth-two history hull.

For a bounded planar field Q and a graph $H : \mathbb{R}^2 \to E_N$, let

$$\mathcal{I}_{Q,H}(u)(\vartheta) = \big(\Phi_Q^\vartheta u, H(\Phi_Q^\vartheta u)\big), \qquad -\Theta_* \le \vartheta \le 0.$$

For each target δ, freeze this preparation at $S = S_\delta$, introduce a dummy amplitude $\rho \in [-\delta, \delta]$, and recover the physical chart at $\rho = \delta$. Writing the prepared version of equations (58) and (60) as $u' = q_0(u) + \rho F_N$, $\rho h' = A_N h + \rho G_N$, its special-flow transform is

$$\mathcal{T}_{Q,N}(Q, H)(u) = q_0(u) + \rho F_N\big(u, H(u), \mathbf{M}_N \mathcal{I}_{Q,H}(u); \rho, \nu, \zeta\big), \tag{67}$$

$$\mathcal{T}_{H,N}(Q,H)(u) = \rho \int_0^\infty e^{A_N r} G_N\big(\mathcal{E}_{Q,H}(\Phi_Q^{-\rho r} u); \rho, \nu, \zeta\big)\ \mathrm{d}r. \tag{68}$$

Here $\mathcal{E}_{Q,H}$ denotes the current state and the evaluation tuple in equation (63). A reduced-flow variation over a time $\Theta_* + \delta r$, followed by equations (61) and (63), gives the common majorant

$$C\delta P(S_\delta) e^{cS_\delta} \int_0^\infty (1 + r^M) e^{-\{D\gamma - c\delta(1+S_\delta)\}r}\ \mathrm{d}r = o(1) \tag{69}$$

for the Lipschitz constant of $(\mathcal{T}_{Q,N}, \mathcal{T}_{H,N})$. The same estimate holds for the highest current block after any derivative in the finite list of the lemma; all other terms contain already controlled lower blocks. More explicitly, order

$$J_{a,b,i,j} = D_u^a \partial_\rho^b \partial_\nu^i \partial_\zeta^j, \qquad b \le 3, \quad i \le 1, \quad j \le 2,$$

first by total order and then by parameter order. Faà di Bruno's formula and the flow variational equations put each current block in the form

$$\mathbf{J}\mathcal{T}_N(Z) = B_{\mathbf{J}}(\mathbf{J}_< Z) + \rho\, \mathcal{A}_{\mathbf{J}}(\mathbf{J}_< Z)[\mathbf{J}Z]. \tag{70}$$

If a derivative hits a flow argument, the vector-field jet gains one state slot and loses one parameter derivative, so it precedes the current block; a product of two positive-order jets has two lower total orders. Derivatives of $\mathbf{M}_N$ vanish, and equation (63) bounds every remaining history evaluation. The coefficient of the last term in equation (70) has the same $o(1)$ bound as equation (69). Induction over this finite block list gives invariant jet balls and, for consecutive Picard iterates, recurrences of the form

$$a_{n+1} \le \kappa a_n + C q^n, \qquad \max\{\kappa, q\} < 1,$$

with the same constants for every N. Uniform convergence of a jet and its first derivative, followed by the fundamental theorem of calculus on state and parameter line segments, identifies the limits as genuine mixed derivatives. This proves convergence of the mixed Picard jets, uniformly for $|\rho| \le \delta$, and gives

$$\big\|(I - D\mathcal{T}_N)^{-1}\big\| \le 2 \tag{71}$$

after decreasing δ_*. Taylor's formula in ρ, evaluated at $\rho = \delta$, gives equation (64); the corresponding graph-pair remainder is the Banach-valued integral

$$\widehat{\mathcal{R}}_{3,N}(\rho) = \frac{1}{2} \int_0^1 (1-t)^2 \partial_\rho^3 (Q_{N,t\rho}, H_{N,t\rho})\ \mathrm{d}t.$$

The Q-component of $\widehat{\mathcal{R}}_{3,N}$ is $\mathcal{R}_{3,N}$ in equation (64).

For the pointwise estimate, split the stable convolution at $L \log(1/\delta)$. Decrease δ_* so that $c\delta(1 + S_\delta) < D\gamma/2$, and then choose L, independently of N, so that

$$CP(S_\delta) e^{cS_\delta} \{1 + L\log(1/\delta)\}^M e^{-D\gamma L \log(1/\delta)/2} = O(\delta^6). \tag{72}$$

On the finite part, the additional reduced-flow backtrack is at most $\delta L \log(1/\delta) = o(1)$. The same triangular induction in nested normal tubes now gives equation (65). At $\rho = 0$, one derivative uses one delay-support backtrack; two derivatives use at most two backtracks and the finite semigroup moments $\int_0^\infty r^k e^{A_N r}\, \mathrm{d}r$, $k = 0, 1$, which are bounded by $k!(D\gamma)^{-k-1}$. Thus the first two coefficients use only the declared depth-two uncut hull. Finally, the fixed-point identity in equation (68), split at any time, is the mild stable equation. Present-time evaluation is a left inverse, so the resulting complete-history map is injective and its retained histories solve the unprepared RFDE exactly.

For the trace solve, repeat the graph construction with the frozen target $S_\delta + B$. Extend its reduced field in the normal coordinate by one fixed finite-order extension: it agrees with the graph field on the shrinking retained tube and equals q_0 near the two endpoints $\gamma_0(\pm R)$. The extension uses the unscaled normal distance from the boundary of the shrinking tube, so it introduces no inverse power of δ into the state or parameter estimates. Only the central pieces which bootstrap back into the exact graph are called RFDE histories; the outer planar tails are selection devices.

It remains to solve for the two selected planar traces. Along

$$\gamma_0(s) = \left(-\frac{s}{2\alpha}, \frac{s^2-2}{4\alpha}\right),$$

the linearized operator is

$$L_0(U,V) = (U' - sU - V, V' + U).$$

Put $E(s) = e^{s^2/2}$, $I(s) = \int_0^s e^{t^2/2}\,\mathrm{d}t$, and

$$\tau(s) = (-1, s)^T, \qquad n(s) = (I(s), E(s) - sI(s))^T.$$

These are respectively the tangent and Gaussian normal solutions, with $\det[\tau, n] = -E$. If $\xi = a\tau + bn$ and $L_0\xi = f$, then

$$a' = E^{-1}\{-(E - sI)f_1 + If_2\} =: A_f, \qquad\qquad b' = E^{-1}(sf_1 + f_2) =: B_f. \tag{73}$$

The phase $U(0) = 0$ is exactly $a(0) = 0$. With the no-incoming normal coordinate prescribed at the outer endpoint, the two one-sided inverses are

$$\begin{aligned} a_a(s) &= -\int_s^0 A_f(t)\,\mathrm{d}t, & b_a(s) &= \beta_a + \int_{-R}^s B_f(t)\,\mathrm{d}t, \\ a_r(s) &= \int_0^s A_f(t)\,\mathrm{d}t, & b_r(s) &= \beta_r - \int_s^R B_f(t)\,\mathrm{d}t. \end{aligned} \tag{74}$$

The elementary estimates

$$|I(s)| \le CE(s)\langle s\rangle^{-1}, \qquad |E(s) - sI(s)| \le CE(s)$$

and the one-sided Gaussian tail bounds show that, for fixed c_0, M_0,

$$\|\mathcal{G}_R^\sigma(f, \beta_\sigma)\|_{c_0, M_0+d} \le C\left(\|f\|_{c_0,M_0} + e^{R^2/2 - c_0 R}\langle R\rangle^{-M_0}|\beta_\sigma|\right), \tag{75}$$

where $d \le 4$, C is independent of R, and $\|g\|_{c,M} = \sup e^{-c|s|}\langle s\rangle^{-M}|g(s)|$. Thus the Gaussian growth is paired with its one-sided integral before a norm is taken.

The tail equation $\mathscr{H}_\alpha = 0$ is not an assumed stable manifold. At an endpoint it is exactly

$$E(s)b = \alpha\{-a + I(s)b\}^2, \tag{76}$$

so the implicit-function theorem solves $b = \mathcal{B}_\sigma(a)$ with zero value and derivative at the origin, uniformly in the normalized boundary norm of equation (75). For $z = \gamma_0 + \xi$, the nonlinear trace equation has the form $L_0\xi = N_{\delta,\nu,\zeta}(s, \xi)$, with

$$|N(s,0)| \le C\delta e^{c|s|}\langle s\rangle^M, \qquad |D_\xi N(s,\xi)| \le C\{\delta e^{c|s|}\langle s\rangle^M + |\xi|\}.$$

Use two tame weights separated by the finite loss d in equation (75). On a ball of radius $C\delta$, the one-sided fixed-point map defined by equations (74) and (76) has Lipschitz factor

$$C\delta e^{cR}\langle R\rangle^{M} = o(1), \qquad R = S_\delta + B. \tag{77}$$

It therefore has a unique fixed point, and its linearization has inverse norm at most two. The ν-source is $O(\delta)$; atomwise row neutrality makes the first ζ-source $O(\delta^2)$. Differentiating the same fixed-point equation proves equation (66) and the stated mixed bounds.

The graph is solved first and the trace equations second. Their combined derivative is block lower triangular; equation (71) and the last trace inverse therefore give a uniform right inverse for the full graph–trace–phase–endpoint range problem. Finally, in the normal coordinate one has the exact identity

$$d(\gamma_0(s) + (U, V)) = V + sU - \alpha U^2.$$

The first estimate in equation (66) makes this $O(\delta e^{cS_\delta}\langle S_\delta\rangle^{M+1})$, which is $o(\delta^\kappa)$ for every fixed $\kappa < 1$. Hence the central trace and all of its retained delay backtracks lie inside the uncut shrinking graph, completing the exact-history assertion. □

The range assertion in theorem 3.1 is the one required by this selected object: the invariant-graph residual followed by its tangent one-sided trace problem. It is not an inverse for arbitrary forcing off the graph in the ambient RFDE history space, and no such stronger inverse is used below.

3.4.2 The mixed curvature return and the normalized gap

Proof of theorem 2.4. We calculate the coefficient after, rather than before, the range problem has been solved. Let $H_{1,N}$ be the first stable graph coefficient in the expansion furnished by theorem 3.1. On the singular orbit,

$$X_0(s) - X_0(s - \theta_k) = -\frac{\theta_k}{2\alpha}. \tag{78}$$

Differentiating the order-δ stable graph equation and using equation (78) gives

$$A_N D_\zeta H_{1,N}(\gamma_0(s)) - \frac{K}{2\alpha} P_{\perp,N}\dot{M}_{1,N}\mathbf{1} = 0.$$

Thus

$$D_\zeta H_{1,N}(\gamma_0(s)) = h_{*,N}, \qquad h_{*,N} = \frac{K}{2\alpha} A_N^{-1} P_{\perp,N}\dot{M}_{1,N}\mathbf{1}. \tag{79}$$

This vector is independent of s. Put

$$r_N = \pi_N^T \operatorname{diag}(c_N) h_{*,N}. \tag{80}$$

Because $\pi_N^T R_{k,N} = 0$ as a row identity, the structural derivative of $\pi_N^T \mathcal{L}_{N,\zeta}[\phi]$ vanishes for every vector history ϕ. It therefore removes the direct critical delay term, the translated history $g_N X^2$, and the flow-history corrections before any moment reduction is made. The translation is independent of ζ. The only order-δ^2 structural term left in equation (58) is $-2X\pi_N^T \operatorname{diag}(c_N)h_{*,N}$. Consequently

$$D_\zeta q_{1,N} = 0, \qquad D_\zeta q_{2,N}(\gamma_0(s), \nu, 0) = \Big(\frac{r_N}{\alpha}s, 0\Big)^T. \tag{81}$$

This proves both the cancellation of the lower-order response and the mixed-curvature return; it is an identity for the actual graph jets of equation (64), not a formal current-state expansion.

We next include the trace and endpoint contributions. The normalized adjoint and the first-integral differential along γ_0 are

$$\psi(s) = e^{-s^2/2}(s,1)^T, \qquad \nabla\mathscr{H}_\alpha(\gamma_0(s)) = \frac{\alpha e}{2}\psi(s). \tag{82}$$

The first reduced coefficient is

$$q_{1,N,X} = -\kappa_N X^3 + K\pi_N^T \mathcal{L}_{N,0}[\mathbf{1}X], \qquad q_{1,N,Y} = \nu. \tag{83}$$

Along γ_0, the delay term in equation (83) is constant in s, by equation (78), and hence pairs to zero with the odd function $se^{-s^2/2}$. Using $\int e^{-s^2/2}\,\mathrm{d}s = \sqrt{2\pi}$, $\int s^2 e^{-s^2/2}\,\mathrm{d}s = \sqrt{2\pi}$, and $\int s^4 e^{-s^2/2}\,\mathrm{d}s = 3\sqrt{2\pi}$, we obtain

$$\int_{\mathbb{R}} \psi^T q_{1,N}(\gamma_0(s),\nu)\,\mathrm{d}s = \sqrt{2\pi}\left(\nu + \frac{3\kappa_N}{8\alpha^3}\right) = \sqrt{2\pi}(\nu - \nu_{0,N}), \tag{84}$$

$$\int_{\mathbb{R}} \psi^T D_\zeta q_{2,N}(\gamma_0(s),\nu,0)\,\mathrm{d}s = \sqrt{2\pi}\frac{r_N}{\alpha}. \tag{85}$$

For clarity, these whole-line integrals do not replace a finite-section boundary-value problem. The two endpoint values of $\mathscr{H}_\alpha$ are zero by construction, and integrating $\mathrm{d}\mathscr{H}_\alpha/\mathrm{d}s$ along the two one-sided traces includes the phase, both endpoints, and the moving complete histories. The Green bounds in theorem 3.1 and equation (65) allow the traces to be replaced by γ_0 in the leading terms. The omitted Gaussian tails and the difference between two preparations in the declared class are bounded, with the same parameter derivatives, by

$$Ce^{-S_\delta^2/2+cS_\delta}\langle S_\delta\rangle^M = C\delta^{p-o(1)}. \tag{86}$$

Choosing the fixed p after the finite tame losses makes this smaller than the algebraic remainders below. The pairings in equations (84) and (85) therefore yield, uniformly in N, in the preparation class, and on the common parameter box,

$$\frac{\mathscr{D}_{N,\mathcal{P}_N}}{\delta} = \sqrt{2\pi}(\nu - \nu_{0,N}) + O(\delta), \tag{87}$$

$$\partial_\nu \mathscr{D}_{N,\mathcal{P}_N} = \delta\sqrt{2\pi} + O(\delta^2), \tag{88}$$

$$\partial_\zeta \mathscr{D}_{N,\mathcal{P}_N} = \delta^2\sqrt{2\pi}\frac{r_N}{\alpha} + O(\delta^3 + \delta^2|\zeta|), \tag{89}$$

$$\partial_{\zeta\zeta} \mathscr{D}_{N,\mathcal{P}_N} = O(\delta^2). \tag{90}$$

Thus the graph, trace, endpoint, and remainder estimates needed for the root are all part of the same uniform construction.

Choose a common $c_0 > 0$ smaller than the boundary distance of the $\nu_{0,N}$ from ∂I_ν. Equations (87) and (88) give opposite endpoint signs and strict monotonicity on $|\nu - \nu_{0,N}| < c_0$ for small δ, ζ, proving the unique root in equation (30). Along that root,

$$\partial_\zeta \nu_{c,N,\mathcal{P}_N} = -\delta\frac{r_N}{\alpha} + O(\delta^2 + \delta|\zeta|). \tag{91}$$

Integrating equation (91) from 0 to ζ and multiplying by δ^2 gives equations (31) and (32), with a constant independent of N and of the declared preparation. On the section $X = 0$, the map $Y \mapsto \mathscr{H}_\alpha(0,Y)$ has derivative $\alpha e/2 \neq 0$ at the singular crossing. Hence a zero of the gap is equality of the two reduced current states. Injectivity of the history graph in theorem 3.1 then makes it equality of the two retained complete RFDE histories. The common value of α and equation (29) also give $\inf_N |\mathscr{C}_N| = \alpha^{-1}\inf_N |r_N| > 0$, as asserted. This completes the proof of theorem 2.4. □

3.4.3 A uniform nonzero family without a synchrony quotient

It remains to verify that the nondegeneracy hypothesis is nonempty without replicating a lower-dimensional synchrony class. Use the explicit profile stated after theorem 2.4. Direct summation gives

$$\pi_N^T s_N = 0, \qquad \pi_N^T s_N^{\circ 2} = 1, \qquad \|s_N\|_\infty = \sqrt{\frac{3(N-1)}{N+1}} < \sqrt{3}, \tag{92}$$

and its entries are pairwise distinct. Thus $c_N = \mathbf{1} + \sigma s_N$ is positive for $0 < \sigma < 1/\sqrt{3}$. Moreover,

$$\left\| s_N \pi_N^T x \right\|_N \leq 2\sqrt{3}\ \|x\|_N\,,$$

so the layer bound is uniform. Since $B_{k,N} \pm \zeta s_N \pi_N^T$ has entries $(b \pm \zeta s_{i,N})/N$, both perturbed layers are positive throughout the common interval $|\zeta| < b/\sqrt{3}$.

Here $A_N = -D\mathrm{I}$ on E_N and $\dot{M}_{1,N}\mathbf{1} = (\theta_0 - \theta_1)s_N$. Together, equations (79), (80) and (92) give

$$r_N = -\frac{K\sigma(\theta_0 - \theta_1)}{2D}, \qquad \mathscr{C}_N = \frac{K\sigma(\theta_0 - \theta_1)}{2D} \neq 0, \tag{93}$$

the same number for every $N \geq 2$.

Finally, suppose a synchrony polydiagonal had a cell containing distinct nodes $i \neq j$. Test invariance on a time-constant history with $v_i = v_j \neq 1$. The balanced delayed term vanishes, the two rows of $P_{c,N}$ give identical instantaneous coupling, and the shared resource and cubic terms agree at the two nodes. Their quadratic terms differ by $(c_{i,N} - c_{j,N})(v_i - 1)^2 \neq 0$. The vector field is therefore not tangent to that polydiagonal. Since all curvatures are distinct, only the singleton partition survives. Thus equation (93) is a genuine transverse heterogeneous-network response and not a lifted two-node root.

3.5 Nonlinear synchronization and collective defect

The same diameter estimate applies to the nonlinear network while all node voltages remain in the strip. With M_0 the retained-history diameter at entrance, the validated nonlinear comparison gives

$$M(t) \leq M_0 e^{-(t-t_0)/10}. \tag{94}$$

Let $(\bar{v}, \bar{w}) = (\pi^T v, \pi^T w)$ and write $\pi^T B_j = \frac{1}{2}\pi^T + b_j^T$. Row mass makes $b_j^T \mathbf{1} = 0$, so oscillation duality controls the delayed left-imbalance. The collective scalar equation has a forcing satisfying

$$\begin{aligned} |R_{\mathrm{coll}}(t)| \leq{}& \frac{391}{20000}\{\delta_0 M(t-\tau_0) + \delta_1 M(t-\tau_1)\} \\ &+ \frac{1403}{400} M(t)^2 + \frac{3}{800}\{M(t-\tau_0)^2 + M(t-\tau_1)^2\}. \end{aligned} \tag{95}$$

With $\mathcal{H}_M(t) = \sup_{t-r \leq s \leq t} M(s)$, this gives

$$|R_{\mathrm{coll}}(t)| \leq \frac{391}{20000}\delta_B \mathcal{H}_M(t) + \frac{703}{200}\mathcal{H}_M(t)^2. \tag{96}$$

Retaining the prescribed history for the full delayed residence intervals gives the resolved forward bound

$$\|R_{\mathrm{coll}}\|_{L^1(t_0,\infty)} \leq \frac{391}{20000}\{\delta_0(10 + 4\sqrt{5}) + \delta_1(10 + 5\sqrt{5})\}M_0$$

$$+\left(\frac{703}{40}+\frac{27\sqrt{5}}{800}\right)M_0^2, \tag{97}$$

and, when only δ_B is retained,

$$\|R_{\mathrm{coll}}\|_{L^1(t_0,\infty)} \le \frac{391(2+\sqrt{5})}{4000}\delta_B M_0 + \frac{56483}{3200}M_0^2. \tag{98}$$

These constants are independent of N and of the admitted topology. The term $27\sqrt{5}/800$ is the exact residence correction for the two delayed quadratic remainders, while the factors $10+\tau_j$ retain the corresponding delayed linear histories. They cannot be removed for an arbitrary retained history.

For the left-balanced subclass $\delta_B = 0$, every linear collective term cancels and equation (95) reduces to

$$|R_{\mathrm{coll}}(t)| \le \frac{1403}{400}M(t)^2 + \frac{3}{800}\{M(t-\tau_0)^2 + M(t-\tau_1)^2\},$$

and hence, with $\mathcal{H}_M(t) = \sup_{t-r\le s\le t} M(s)$,

$$|R_{\mathrm{coll}}(t)| \le \frac{703}{200}\mathcal{H}_M(t)^2,$$

$$\|R_{\mathrm{coll}}\|_{L^1(t_0,\infty)} \le \left(\frac{703}{40}+\frac{27\sqrt{5}}{800}\right)M_0^2 \le \frac{56483}{3200}M_0^2. \tag{99}$$

Thus the proved uniform estimate for the enlarged class contains a resolved term linear in M_0. Without an additional cancellation, the present argument does not yield a purely quadratic enlarged-class estimate.

None of these estimates is, by itself, a basin theorem. In the balanced subclass, if a scalar route accepts entrance error δ_{mean} and additive forcing whenever

$$L_0\delta_{\mathrm{mean}} + \|e\|_{L^1} < \eta_{\mathrm{route}},$$

and if d_{strip} and d_{lift} are respectively a scalar strip margin and a product-basin lift margin, then the exact sufficient network budget is

$$R_0 < \min\{d_{\mathrm{strip}}, d_{\mathrm{lift}}\}, \qquad L_0R_0 + \frac{56483}{3200}R_0^2 < \eta_{\mathrm{route}}. \tag{100}$$

Here $R_0 = \max\{\delta_{\mathrm{mean}}, M_0\}$. A first-strip-exit contradiction proves that the hypotheses bootstrap until routing is complete. The scalar route, strip, lift, and gap constants in equation (100) remain model-specific release data. No concrete radius has been validated, and the released nonbalanced theorem does not promote an asynchronous threshold or basin.

3.6 Scope of the network theorem

The leaky transverse and triangular-transfer theorem allows arbitrary finite size, a nonnegative row-stochastic current layer with $\tau(Q) \le 1/2$, and fixed nonnegative delayed layers of row mass $1/2$. It does not require common delay-layer left balance. It remains conditional on the scalar phase-fixed root and collective preparation, and it does not include signed coupling, moving delay support, closing Dobrushin gaps, heterogeneous node vector fields, an invariant asynchronous strip, or physical onset.

The separate shared-resource structural-root theorem allows heterogeneous curvature under a common Dobrushin gap, fixed delay support, an atomwise row-neutral structural direction, and the bounded admissible preparation class stated in theorem 2.4. The graph and selected-trace range solves for that class are constructed in theorem 3.1; they are not extra hypotheses. The exact finite-δ root remains preparation indexed. The theorem does not cover order-one changes of the base generator, moving delay atoms, a closing mixing gap, a preparation-independent root, or identification with an independently selected physical outer canard. These are mathematical boundaries, not numerical omissions.

4 Selected pulse thresholds and conditional control coordinates

We now turn from a selected complete-history root to the physical pulse experiment. The two objects are kept separate. A selected pulse/stable-sheet threshold is an intersection in the RFDE history space; it becomes a biological onset, and its signed offset becomes a safety coordinate, only after two-sided basin routing has been proved.

4.1 The inner section and its stable sheet

Fix the phase-zero voltage section through the inner periodic orbit Γ_i. The exact orbit crosses this section with speed bounded below by 0.2067539137. After removing the translation multiplier, the section return has one simple real unstable multiplier and a stable complement. The validated characteristic root satisfies

$$0.6983604100 < s_u < 0.6983604300, \tag{101}$$

while the stable spectral radius is at most $e^{-0.01}$.

Let q be the unstable right column and let $\widetilde{f}$ be the raw left functional supplied by the bordered problem. Once $\widetilde{f}(q) \neq 0$, put $f = \widetilde{f}/\widetilde{f}(q)$. Write the section coordinates as

$$u = f(\phi), \qquad s = \phi - qf(\phi).$$

A forward Lyapunov–Perron construction seeks the local stable sheet in the form

$$u = \psi(s), \qquad \psi(0) = D\psi(0) = 0. \tag{102}$$

The graph transform is estimated in six directed blocks corresponding to stable and unstable inputs and outputs. Its difficult term is the stable part of the second return variation in two unstable directions,

$$Y_{qq} - qf(Y_{qq}).$$

Bounding Y_{qq} and $qf(Y_{qq})$ separately loses the cancellation and does not close the graph radius.

4.2 The continuous RFDE adjoint functional

For a discrete-delay variational equation

$$\dot{z}(t) = A_0(t)z(t) + \sum_j A_j(t)z(t - \tau_j),$$

the periodic advanced adjoint carries a current atom and an absolutely continuous history density. Its bilinear history pairing, not a Hermitian inner product, defines the raw section functional $\widetilde{f}$. The Fourier cokernel-to-history reconstruction has the exact reversal

$$\widehat{r}_n = E_{-,-n}, \tag{103}$$

with no complex conjugation. This distinction is essential for the annihilation of the neutral tangent.

The directed tail solve proves

$$\text{tail contraction} < 0.104433459418, \qquad \|r_{\text{tail}}\|_{\ell^1} < 0.01150531427, \qquad \|r\|_{\ell^1} < 0.1835103672. \tag{104}$$

The Grushin border derivative and the phase-averaged RFDE bilinear identity then give the directed normalization

$$0.000305331752871 < \left|\widetilde{f}(q)\right| < 0.000546927481134. \tag{105}$$

Thus the Fourier cokernel, continuous periodic adjoint, and Route–C history functional can be normalized to the same object f. A recovery-only history shard has normalized action greater than 12.9099, which also rules out any current-voltage-only projection.

For a 240-step finite-section diagnostic, performing the correlated deflation before taking the norm gives

$$\|Y_{qq} - qf_N(Y_{qq})\|_\infty = 7.26112 < 12, \tag{106}$$

whereas the separate triangle estimate is 56.7830. This is strong evidence for the required correlation, but equation (106) itself is not interval evidence.

Proposition 4.1 (Base-orbit correlated uu block)**.** *Let q^Σ be the physical Route–C section unstable direction and let Y_{qq} be the physical-time second returned variation at the validated center inner orbit. Then*

$$\begin{aligned}
\left\|V_{qq} - \widehat{V}_{qq}\right\|_P &\le 0.00808459995104, \\
\left|\widetilde{f}(q^\Sigma)\right| &\ge 4.2563852678 \times 10^{-4}, \\
\left\|Y_{qq} - q^\Sigma \widetilde{f}(Y_{qq})/\widetilde{f}(q^\Sigma)\right\|_Y &\le 0.0475498854764, \\
\left\|q^\Sigma\right\|_Y &\ge 0.0775543158981.
\end{aligned} \tag{107}$$

Consequently the normalized stable-output block satisfies

$$C^{uu}_{s,\text{base}} \le \frac{0.0475498854764}{(0.0775543158981)^2} < 7.905649079 < 12. \tag{108}$$

The proof uses 1042 physical method-of-steps cells with both delays aligned and a degree-24, 192-bit Taylor–Bernstein residual. It defines the quotient through the same-row identity

$$\frac{\widetilde{f}(Y_{qq})}{\widetilde{f}(q^\Sigma)} - c = \frac{\widetilde{f}(Y_{qq} - cq^\Sigma)}{\widetilde{f}(q^\Sigma)}$$

and applies the adjoint error only after the center history has been deflated. A direct atom–density action on the polynomial guide, the advanced covariance $e^s T$, continuous voltage/recovery source residuals, all cell seams, the density-product variation, and the normalization defect give a total correlated action error at most $1.500297667 \times 10^{-6}$. In particular, no exact inhomogeneous adjoint identity is silently assigned to the approximate Fourier row.

Theorem 4.1 closes the former covector/cancellation bottleneck only at one periodic base orbit. Its uniform split-ball inflation and the other five blocks—hence all six uniform split-ball Hessian blocks—the split nonlinear return tube, and the first-positive-return/no-earlier-hit enclosure remain unvalidated. The discrete stable-power constant is supplied separately by theorem 4.3. Accordingly, the six-block quantitative stable graph equation (102) remains open; a qualitative local stable graph for the selected coordinate map is proved below in theorem 4.11.

The exact inflation budget shows what a uniform proof must achieve. Since the base value leaves margin 4.0943509217, it is sufficient to prove a split-ball Lipschitz constant strictly below

$$L_{uu}^{\mathrm{req}} = 2408.4417186722.$$

A source-bound 192-bit scalar P-logarithmic-norm attempt closes the self-consistent enclosure on all 1042 physical cells, but first loses the radius-0.01 local section ball on cell 581, where its voltage radius is already 0.010059402; its period-end bound is 2.151313. Thus this positive scalar majorant is rigorously rejected as the uniform inflation route. It does not rule out a correlated bound. Indeed,

$$2\tau_0 < T < \tau_0 + \tau_1 < 3\tau_0$$

reduces the signed linear ingress to only the four Volterra words $\varnothing, (\tau_0), (\tau_1), (\tau_0, \tau_0)$. The required replacement is to form $U(t)P_s = U(t) - U(t)qf/f(q)$ across those words before taking total variation.

Proposition 4.2 (Primitive ingress and physical-frame obstruction)**.** *On the* 1042*-cell delay-aligned physical grid, the degree-24,* 192*-bit Taylor–Bernstein residuals of the five primitives* F, G, C_0, C_1, C_{00} *are bounded above respectively by*

$$7.30427 \times 10^{-7}, \quad 2.54960 \times 10^{-7}, \quad 3.33270 \times 10^{-7}, \quad 2.56054 \times 10^{-7}, \quad 3.06759 \times 10^{-11}.$$

After every Hermite seam and coefficient tail is included, the moving-frame maximum error radii are at most

$$4.92742 \times 10^{-4}, \quad 2.42414 \times 10^{-4}, \quad 1.73778 \times 10^{-6}, \quad 1.43306 \times 10^{-6}, \quad 3.64819 \times 10^{-11}.$$

The resulting directed primitive ingress into the event-row estimate is at most

$$1.463546184 \times 10^{-3}. \tag{109}$$

By contrast, the declared unprojected scalar physical-frame Gronwall propagation returns F*- and* G*-error radii larger than* 3.04997×10^7 *and* 1.89994×10^7*, respectively, and therefore cannot certify the stable row.*

Proposition 4.3 (Selected terminal stable row)**.** *Let*

$$Y = C([-\tau_{\max}, 0], \mathbb{R}) \times \mathbb{R}, \qquad \Sigma_0 = \{h \in Y : h_v(0) = 0\},$$

and set

$$A = \Pi_T U(T,0)|_{\Sigma_0}, \qquad f = \widetilde{f}/\widetilde{f}(q^\Sigma), \qquad P_s = \mathrm{I} - q^\Sigma f, \qquad E_s = \ker f.$$

Here $\Pi_T y = y - \dot{X}_T y_v(0)/\dot{v}(T)$ *is the terminal section projection. For the selected near-one-period phase-fixed linear map,*

$$\|AP_s\|_{\Sigma_0 \to Y} \le 0.009896427481610001 < 0.1. \tag{110}$$

Moreover,

$$AP_s = P_s A = P_s A P_s, \qquad AP_s(\Sigma_0) \subset E_s.$$

Consequently, for $A_s = A|_{E_s}$*,*

$$\|A_s^n\|_{E_s \to Y} \le 0.1^n, \qquad K_s = 1 \quad (n \ge 1). \tag{111}$$

The certificate forms the common signed row $R_\theta \Pi_T U(T,0)(\mathrm{I}-q^\Sigma f)$ before every modulus. The exact section quotient removes the current-voltage δ_0 atom, while the current-recovery atom remains active. The true-period enclosure gives $T>\tau_{\max}$, so the complete returned history lies after time zero and has no unadvanced identity block. Its only method-of-steps words are $\varnothing,(\tau_0),(\tau_1),(\tau_0,\tau_0)$. The continuous calculation covers 641 returned-history phase rows plus one recovery-output row against 640 input-history cells, hence $(641+1)\times 640 = 410{,}880$ support rectangles. The independently outward-rounded ledger is

$$
\begin{aligned}
E_{\mathrm{ctr}} &\le 0.004362999710080374, & E_{\mathrm{bin}} &\le 0.000010000000000001,\\
E_{4\mathrm{I}} &\le 0.001463546183857267, & E_{\mathrm{rank}} &\le 0.004059226965383500,\\
E_{\mathrm{ratio}} &\le 6.506172913144082\times 10^{-7}, & E_T &\le 4.004997545987388\times 10^{-9}.
\end{aligned}
$$

The displayed terms sum to at most 0.009896427481610003; the unrounded directed calculation gives the sharper bound in equation (110). The Stage–4I primitive tubes enter only through their direct algebraic image: neither the unknown terminal norm nor K_s is used in its own estimate.

Theorem 4.2 is therefore only the primitive parent budget. Theorem 4.3 supplies the missing common terminal row, true-T_+ coverage, section quotient, and continuous output-phase supremum, and thereby closes the discrete one-step stable-map norm.

A stronger intermediate-time stable-flow certificate could instead be obtained without assuming the stable-flow constant that it is meant to prove. Let $U_s(t,s)=U(t,s)P_s(s)$, and let $\widehat{U}_s$ be one common signed polynomial guide whose initial defect and complete method-of-steps residual lie in the exact transported stable bundle. The latter includes the differential residual, history-transport boundary, delay activations, and cell seams. The complete-history norm also retains the unadvanced translation/identity block; it is not only a norm of the propagated density. For $\alpha \ge 0$, put

$$
K=\sup_{s\le t} e^{\alpha(t-s)}\left\|U_s(t,s)\right\|, \qquad \widehat{K}=\sup_{s\le t} e^{\alpha(t-s)}\left\|\widehat{U}_s(t,s)\right\|,
$$

$$
\Delta=\sup_s \|D_s\| + \sup_{s\le t}\int_s^t e^{\alpha(r-s)}\left\|R(r,s)\right\|\,dr.
$$

Duhamel's formula remains in the stable bundle and yields

$$
K\le \widehat{K}+K\Delta, \qquad \Delta<1 \Longrightarrow K\le K_{\mathrm{int}} := \frac{\widehat{K}}{1-\Delta}. \tag{112}
$$

Let $\widehat{M_s}$ be the corresponding terminal event-corrected row, let $C_{\Pi,T}$ bound the exact event correction, let Δ_T be the preprojection endpoint residual budget, and let $\varepsilon_{\Pi,T}$ contain the remaining event-row and speed-denominator errors. Then

$$
\|M_s\| \le \left\|\widehat{M_s}\right\| + C_{\Pi,T}K_{\mathrm{int}}\Delta_T + \varepsilon_{\Pi,T}. \tag{113}
$$

The event projection cannot be assigned norm one; its sampled factor is about 2.01358. A strict right side below $\rho_s<1$ would provide the corresponding intermediate-time stable-flow estimate. The exact projection is essential: propagating the five primitives separately violates the stable-residual premise and recovers the rejected unstable growth. The phase correction is applied only at the terminal section; a moving Π_t would add a derivative term to the residual equation. This route is not used by theorem 4.3, whose fixed-start terminal calculation already proves the discrete power estimate. At present the common full (s,t) projected residual, Δ, and $C_{\Pi,T},\Delta_T,\varepsilon_{\Pi,T}$ remain unvalidated. Thus equations (112) and (113) are an optional route toward intermediate-flow, nonlinear-tube, or no-earlier-hit control, not a prerequisite for $K_s=1$.

4.3 Eventually smooth event Hessian and two-return graph budget

The discrete estimate in theorem 4.3 does not by itself make the moving complete-history return C^2 on a ball of arbitrary continuous histories. We first record the exact event-aligned Hessian, so that the remaining analytic and numerical gates are explicit.

Let $S_t(x)$ be the fixed-time complete-history solution map, let $T(x)$ be a selected crossing of the phase section, and put

$$a(x) = \dot{v}_x(T(x)), \qquad \ell_0(y) = y_v(0).$$

For initial directions h, k, let U_h and V_{hk} be the first and second fixed-time variations, with $V_{hk,0} = 0$, and use the superscript T for the complete segment translated to the selected event. Set

$$n_h = \ell_0(U_h^T), \qquad\qquad T_h = -\frac{n_h}{a}, \tag{114}$$

$$Z_{hk} = V_{hk}^T - \frac{\dot{U}_h^T n_k + \dot{U}_k^T n_h}{a} + \frac{\ddot{X}_T n_h n_k}{a^2}, \qquad\qquad T_{hk} = -\frac{\ell_0(Z_{hk})}{a}. \tag{115}$$

Define the moving event projection

$$\Pi_x y = y - \dot{X}_T \frac{\ell_0(y)}{a(x)}. \tag{116}$$

Proposition 4.4 (Selected-return Hessian)**.** *On any common C^2 selected-event tube on which $a(x) > 0$, the moving complete-history return $P(x) = S_{T(x)}(x)$ satisfies*

$$DP(x)h = \Pi_x U_h^T, \qquad D^2P(x)[h,k] = \Pi_x Z_{hk}. \tag{117}$$

The phase projection in equation (117) *is applied exactly once. It precedes the fixed spectral deflation P_s or the fixed unstable row $\widehat{f}$; those two projections are not interchangeable.*

Proof. Differentiate the scalar event identity twice to obtain equations (114) and (115), including every terminal-history translation term. Substitution into the second derivative of $S_{T(x)}(x)$ gives equation (117). In particular, $\ell_0(DP\,h) = \ell_0(D^2P[h,k]) = 0$, which also checks that no event correction has been duplicated. $\square$

For the one-period center return, $T > \tau_{\max}$ removes the unadvanced identity block from the *linear* output, but $T < 2\tau_{\max}$. The old one-period argument therefore does not satisfy the C^2 complete- segment condition in theorem 2.5. The selected near-two-period center does: the frozen orbit intervals give

$$T_- - 2\tau_{\max,+} > 14.010740123253945. \tag{118}$$

This proves the center-orbit margin. The next proposition closes the common full-ball event inequality on an explicit, deliberately microscopic scale.

Proposition 4.5 (Explicit common near-two-period event tube)**.** *Let*

$$B_\lambda = \{(x_s, x_u) : \|x_s\|_Y \le 0.0097\lambda,\ |x_u| \le 0.00025\lambda\}, \qquad \lambda_0 = 9 \times 10^{-31}.$$

There is an open reduced-history neighborhood $W_{\text{open}} \subset Y$ and an open coordinate restriction D_{open} containing B_{λ_0} on which every injected history, including every continuous reduced stable history satisfying the displayed norm bound, has one unique positively oriented voltage event in the common physical-time window

$$I_2 = [36.37141989825184198 57,\ 36.37341989825984198 58]. \tag{119}$$

The event time and the complete reduced-history hit are C^2; the initial history and the hit both lie in the fixed radius-0.01 local phase-zero section patch. Uniformly on the larger open comparison scale 9.1×10^{-31}, the endpoint gap is greater than 1.03751×10^{-4}, the event speed is greater than 0.244880, the returned-history radius is less than 5.09376×10^{-4}, and $T_- - 2\tau_{\max} > 14.010740$.

Writing χ for the fixed section chart, the induced map has type $P_{\rm sel} : D_{\rm open} \to D_{\rm out}$, where $D_{\rm out} = \chi(\Sigma_{\rm loc})$. No containment $D_{\rm out} \subset D_{\rm open}$ is asserted.

At the center, $T(Y_*) = 2P$ and the hit is exactly Y_*. The result is neither a self-map of B_{λ_0}, nor a certificate at $\lambda = 1$, nor an event-ordinal statement.

Proof. Write $B = \sup_t |v_*(t) - 1|$. On an orbit-centered tube of radius β, the corrected fast-field row gives

$$H_\beta = 2(1 + B + \beta) + 12\varepsilon\kappa_3(B + \beta), \qquad L_\beta = \max\{A_0 + H_\beta\beta, 2\varepsilon\}.$$

Choose β to be half the certified center endpoint gap and propagate the single complete-history majorant

$$M(t) \le \rho_{\rm open} + \int_0^t L_\beta M(s)\,ds \qquad (0 \le t \le T_+).$$

It retains the arbitrary initial-history translates before each delay activates and covers all later method-of-steps seams. Directed evaluation gives $0.00995(9.1 \times 10^{-31})e^{L_\beta T_+} < \beta$; the corresponding open-scale ceiling is greater than $9.13306649 \times 10^{-31}$. A first-exit argument therefore proves the common flow tube. The center endpoint signs and the Lipschitz perturbation bound give the stated strict signs and speed, so the selected zero is unique in I_2. The complete-history estimate also gives terminal section-patch containment. Finally, apply theorem 2.5 first with the affine compatible lift $\mathcal{I} : W_{\rm open} \to X$, the global polynomial RFDE field, and $g_X = g_Y \circ \pi$. Projection gives the ambient reduced C^2 hit, and restriction through the section injection $j : D_{\rm open} \to W_{\rm open} \cap \Sigma_{\rm loc}$ gives the stated coordinate map. The strict margin in equation (118) supplies the required joint smoothing. □

The preceding scalar comparison is correct but extremely cancellation blind. The recovery equation supplies a weighted energy in which the two instantaneous voltage–recovery cross terms cancel exactly. Exploiting that identity enlarges the same selected-event domain by more than twenty decimal orders.

Proposition 4.6 (Orbit-aware enlarged near-two-period event tube)**.** *With the same anisotropic balls B_λ and the same physical window equation (119), every history in*

$$B_{1.2\times10^{-8}} \tag{120}$$

has one unique positively oriented selected voltage event. The conclusion holds on the strictly larger open scale 1.25×10^{-8}; the event time and complete reduced-history hit are C^2, and the hit remains in the declared radius-0.01 terminal section patch. Uniformly on that open scale, the endpoint gap is greater than 1.05563×10^{-4} and the event speed is greater than 0.206347.

The sufficient construction has scale ceiling

$$\lambda_{\rm en} > 1.2770076307683837 \times 10^{-8}. \tag{121}$$

It proves neither the preferred scale $\lambda = 1$, a same-ball self-map, an event ordinal, any one of the six Hessian caps, nor a pulse crossing.

Proof. Let $x = \eta_v(t)$ and $y = \eta_w(t)$ be the difference from the exact periodic orbit and set

$$z(t)^2 = x(t)^2 + \varepsilon^{-1} y(t)^2.$$

If $a(t)$ is the current variational coefficient and $b_0(t), b_1(t)$ are the two delayed coefficients, then

$$\frac{1}{2}\frac{d}{dt}z^2 = a(t)x^2 - y^2 + \sum_{j=0}^{1} b_j(t)x(t)x(t-\tau_j) + x(t)N(t). \tag{122}$$

The terms $-xy + yx$ have cancelled exactly. For the complete-history running envelope Z, including the untranslated initial voltage history before each delay activates, this gives at every increasing-envelope time

$$D^+Z(t) \le \left[r(t) + \frac{1}{2}H_\beta Z(t)\right] Z(t), \qquad r(t) = \max\{0, \max(a(t), -\varepsilon) + |b_0(t)| + |b_1(t)|\}.$$

Here H_β is the corrected polynomial Hessian row in the proof of theorem 4.5.

On the 1042 delay-aligned cells, 192-bit Taylor–Bernstein arithmetic gives

$$\int_0^{P_{\text{bin}}} r(t)\,dt \le 6.353397548471557, \qquad \sup r(t) \le 0.774979654297297. \tag{123}$$

The enclosure uses the exact binary interval for P_{bin}, rescales by the directed true-period interval, and adds the phase drift between the exact delays $4\sqrt{5}, 5\sqrt{5}$ and their binary centers before bounding the delayed coefficients. Thus neither decimal round-tripping nor a mesh maximum is used as exact data.

For a reduced-Y initial radius ρ, $Z(0) \le \sqrt{1+\varepsilon^{-1}}\rho = \sqrt{6}\rho$. Choose β as half the center endpoint gap. On a hypothetical first exit through $Z = \beta$, integration of the preceding Dini inequality over two true periods and the remaining window gives a complete-history gain at most 332147.926240866. At the open scale 1.25×10^{-8}, the resulting deviation is at most

$$1.011906217152045 \times 10^{-4} < 1.033769568740104 \times 10^{-4} = \beta,$$

a contradiction. The polynomial field is bounded on this closed tube, so continuation through T_+ also holds. The endpoint signs, speed and terminal patch follow from the same deviation bound, and theorem 2.5 supplies the stated C^2 event and complete-history hit. $\square$

Proposition 4.7 (Direct two-period derivative)**.** *Let Q_Y be the direct selected hit map supplied by theorem 4.5 on the reduced physical section, and let*

$$A = \Pi U_Y(P, 0)|_{\Sigma_0}$$

be the proved one-period phase-fixed derivative. Then

$$Q_Y(Y_*) = Y_*, \qquad DQ_Y(Y_*) = A^2. \tag{124}$$

This conclusion does not require a nonlinear one-period return or an identity $Q = P^2$. If $J(x_s, x_u) = x_s + \widehat{q}x_u$ is the section coordinate isomorphism, then

$$DQ_{\text{coord}}(0) = J^{-1}A^2J. \tag{125}$$

For the compatible full-history lift $\mathcal{I}$, the correctly typed identity is

$$DQ_X(\mathcal{I}(Y_*))D\mathcal{I} = D\mathcal{I}\,A^2, \tag{126}$$

not $DQ_X = A^2$.

Proof. Put $M = U_Y(P, 0)$ and $\Pi = I - v\ell/a$, where v is the phase tangent, $\ell = Dg$, and $a = \ell(v)$. Periodicity and the forward variational cocycle give $U_Y(2P, 0) = M^2$ and $Mv = v$. Direct differentiation of the selected event gives $DQ_Y(Y_*) = \Pi M^2|_{\Sigma_0}$. Since $(I - \Pi)y = v\ell(y)/a$ and $\Pi M v = 0$,

$$(\Pi M|_{\Sigma_0})^2 = \Pi M \Pi M|_{\Sigma_0} = \Pi M^2|_{\Sigma_0}.$$

The chart chain rule and the exact lift/projection factorization give equations (125) and (126). □

The two-return route is also quantitatively favorable. Use the fixed unit pair

$$\widehat{q} = q/\left\|q\right\|_Y, \qquad \widehat{f} = \left\|q\right\|_Y f, \qquad P_s = \mathrm{I} - \widehat{q}\widehat{f},$$

the split radii $R_s = 0.0097$, $R_u = 0.00025$, and the desired graph radius $r = 0.0094$. For output $i \in \{s, u\}$, let $C_{i,ss}, C_{i,su}, C_{i,uu}$ bound the six projected blocks of $D^2\mathcal{Q}$, with the event projection formed before P_s or $\widehat{f}$.

Proposition 4.8 (Fixed two-step linear splitting)**.** *Let $A \in \mathcal{L}(\Sigma_0)$, $q \in \Sigma_0$, and $f \in \Sigma_0^*$ satisfy*

$$Aq = \mu_u q, \qquad fA = \mu_u f, \qquad f(q) = 1,$$

and put $P_u = qf$, $P_s = \mathrm{I} - qf$, $E_u = \operatorname{span}\{q\}$, and $E_s = \ker f$. Then these are complementary projections, $\Sigma_0 = E_s \oplus E_u$, and both spaces are invariant under A. For the linear two-step operator $B = A^2$,

$$BP_s = P_sB = P_sBP_s = (AP_s)^2, \qquad BP_u = P_uB = \mu_u^2 P_u. \tag{127}$$

If $\|A_s^n\| \leq K_s\rho_{s,1}^n$ and $\|(A_u)^{-n}\| \leq K_u\rho_{u,1}^n$, then

$$\|B_s^n\| \leq K_s(\rho_{s,1}^2)^n, \qquad \|(B_u)^{-n}\| \leq K_u(\rho_{u,1}^2)^n. \tag{128}$$

For the source-bound inner-orbit operator of theorem 4.3, this gives

$$K_{s,2} = 1, \quad \rho_{s,2} \leq 0.01, \qquad K_{u,2} = 1, \quad \rho_{u,2} \leq 0.302183501335053468766049321269. \tag{129}$$

The last number is an inverse/backward unstable rate; the forward multiplier has modulus $|\mu_u|^2 > 1$.

Proof. The normalization $f(q) = 1$ proves the complementary-projection identities. The two eigenrelations give $AP_s = P_sA = P_sAP_s$ and $AP_u = P_uA = \mu_uP_u$. Squaring gives equation (127), while $B_s^n = A_s^{2n}$ and, on the one-dimensional unstable space, $(B_u)^{-n} = \mu_u^{-2n}\mathrm{I}$, proving equation (128). The numerical values use the proved $K_s = 1$, $\rho_{s,1} \leq 0.1$ and the source-bound $\rho_{u,1} \leq 0.549712198641301273\ldots$. □

This proposition is linear. If a nonlinear selected map $P : D \to \Sigma_0$ is used to define $\mathcal{Q} = P \circ P$, its correct domain is $D_2 = D \cap P^{-1}(D)$; only there does the chain rule give $D\mathcal{Q}(0) = A^2$. Conversely, a direct near-two-period selected map has derivative A^2 by theorem 4.7, without first being represented as P^2. Neither route turns linear invariance into nonlinear invariance of E_s or E_u.

Proposition 4.9 (Conditional two-return graph arithmetic)**.** *Suppose the direct selected near-two-period map extends from the microscopic domain in theorem 4.5 to the declared split ball, is C^2 there, and returns that ball to the graph-transform domain. Identify its coordinates with Σ_0 through J and use the pullback norm $\|z\|_J = \|Jz\|_Y$. Then theorems 4.7 and 4.8 supply $D\mathcal{Q}(0) = B = A^2$, $K_{s,2} = K_{u,2} = 1$ and the stable forward and unstable inverse/backward rates*

$$\rho_{s,2} \leq 0.01, \qquad \rho_{u,2} \leq (0.549712198641302)^2,$$

Take graph weight $\beta = 0.9999$, and suppose the map has the six uniform Hessian bounds

$$(C_{s,ss}, C_{s,su}, C_{s,uu}, C_{u,ss}, C_{u,su}, C_{u,uu}) \le (1, 10, 1000, 5, 10, 1000). \tag{130}$$

Then the exact split Lyapunov–Perron majorant has Perron bound less than 0.193086*, stable and unstable self-map slacks greater than* 1.96408×10^{-4} *and* 1.24109×10^{-4}*, respectively, and the unit-$\widehat{q}$ graph budgets*

$$\left\|\widehat{\psi}\right\| \le 1.25886 \times 10^{-4}, \qquad \left\|D\widehat{\psi}\right\| \le 0.026559. \tag{131}$$

Thus the graph-transform arithmetic closes under the displayed premises. The preferred-ball extension, return-domain containment, and six Hessian bounds remain premises; the center derivative, splitting, and rates do not.

A source-bound but nonrigorous signed finite-section pilot preserves the moving-event quotient, translates the entire returned history at the same event time, applies the fixed Stage–4L $\widehat{q}/\widehat{f}$ pair, and only then takes the six tensor norms. The primary $n_{\text{step}} = 240$ row is approximately

$$(7.45637 \times 10^{-7},\ 4.46663 \times 10^{-3},\ 29.353924, 0.594988,\ 0.573991,\ 158.531687),$$

and the disclosed last-two-mesh heuristic envelope is

$$(2.35230 \times 10^{-6},\ 0.00872968,\ 32.117788, 0.596928,\ 0.577596,\ 158.531779). \tag{132}$$

All six entries lie well inside equation (130). The adapter at the current-history endpoint is a one-sided four-node stencil; an earlier zero-future convention is retained only as a discarded audit. Neither mesh convergence nor the heuristic envelope is an interval error bound, so equation (132) is design evidence and not a hypothesis verification.

The obstruction is not merely that three meshes are too few.

Lemma 4.10 (Finite-sampling obstruction on an arbitrary C^0 ball)**.** *Let K be a nontrivial compact interval, $X = C(K)$ with the supremum norm, and let $S_N h = (h(t_1), \ldots, h(t_N))$ sample any finite node set. For every bounded linear reconstruction $R_N : \mathbb{R}^N \to X$,*

$$\|\mathrm{I} - R_N S_N\|_{\mathcal{L}(X)} \ge 1. \tag{133}$$

Moreover, nodal data cannot determine all unit bilinear forms on X: for any off-grid t_*, the form $B_*(h,k) = h(t_*)k(t_*)y_*$, with $\|y_*\| = 1$, is invisible to inputs whose sampled values vanish.

Proof. Choose $t_* \notin \{t_1, \ldots, t_N\}$ and a continuous spike h with $\|h\|_\infty = 1$, $h(t_j) = 0$, and $h(t_*) = 1$. Then $S_N h = 0$, so $(\mathrm{I} - R_N S_N)h = h$, which proves equation (133). Applying the same spike in both slots proves the bilinear assertion. □

Consequently, the Stage–4Q grid trend cannot be promoted by mesh refinement alone to an operator-norm bound on the arbitrary continuous-history ball. A strict route must instead interpret directed tensor coefficients as physical evaluation atoms and bound the signed atom–density/bimeasure residual to the true RFDE Hessian. If $A = (a_{oij})$ is such a directed finite tensor and the output reconstruction has norm one, its atomic lift satisfies

$$\|B_N\| \le \max_o \sum_{i,j} |a_{oij}|\,. \tag{134}$$

This makes the row-ℓ^1 shape of the pilot appropriate, but supplies no residual bound. The frozen continuous splitting gives $\left\|\widehat{f}\right\| \le 21.810501$ and $\|P_s\| \le 22.810501$; hence an ambient Hessian error can be amplified by as much as $\|P_s\|^3$. The resulting simultaneous raw-error design ceiling is 7.58219×10^{-5}. Direct enclosure of each already projected signed bimeasure is therefore the preferred rigorous route; all six actual block slots remain open.

The preceding obstruction leaves intact the continuous dual structure of the RFDE. The next result both identifies the exact split norm needed by a validated residual calculation and records the qualitative stable manifold that follows without a numerical Hessian cap.

Proposition 4.11 (Split quotient, bimeasure residual, and local selected graph)**.** *Let $\ell_0(\phi, w) = \phi(0)$, choose a Hahn–Banach extension $\bar{f} \in Y^*$ of $\widehat{f} \in \Sigma_0^*$, and write*

$$E_s = \ker \ell_0 \cap \ker \bar{f}, \qquad E_u = \operatorname{span}\{\widehat{q}\}.$$

For every $\mu \in Y^*$, restriction to the stable input space has the exact quotient norm

$$\|\mu|_{E_s}\| = \inf_{\alpha,\beta\in\mathbb{R}} \left\|\mu - \alpha\ell_0 - \beta\bar{f}\right\|_{Y^*}, \qquad \sup_{|c|\le 1} |\mu(c\widehat{q})| = |\mu(\widehat{q})|. \tag{135}$$

Here $Y^* = \mathcal{M}([-\tau_{\max}, 0]) \times \mathbb{R}$, with the total-variation plus recovery-atom norm, and the quotient value is independent of the chosen extension $\bar{f}$. Denote the two input costs in equation (135) by $d_s(\mu)$ and $d_u(\mu)$, respectively, and put

$$\omega_s(y) = \left\|y - \widehat{q}\widehat{f}(y)\right\|_Y, \qquad \omega_u(y) = |\widehat{f}(y)|.$$

Here $I_s : E_s \hookrightarrow \Sigma_0$ and $I_u : \mathbb{R} \to \Sigma_0$, $I_u c = c\widehat{q}$, are the input injections, while $O_s = P_s$ and $O_u = \widehat{f}$ are the output maps. For each $(o, a, b) \in \{s, u\}^3$, let $B_i(z) = O_o D^2 Q_Y(Y_* + Jz)[I_a \cdot, I_b \cdot]$, with the affine section charts understood and the two mixed-input orders identified, giving the six directed blocks indexed by i. If a signed atom–density bimeasure has an absolutely summable or Bochner-integrable representation $\mathcal{B} = \sum_r y_r \otimes \mu_r \otimes \nu_r$, with $y_r \in \Sigma_0$, define

$$\mathcal{N}^o_{ab}(\mathcal{B}) = \inf_{\mathcal{B}=\sum_r y_r\otimes\mu_r\otimes\nu_r} \sum_r \omega_o(y_r) d_a(\mu_r) d_b(\nu_r), \qquad a, b, o \in \{s, u\}.$$

Then

$$\|O_o\mathcal{B}[I_a \cdot, I_b \cdot]\|_{\mathrm{bil}} \le \mathcal{N}^o_{ab}(\mathcal{B}). \tag{136}$$

Consequently, if C_i is an outward split-projective center bound and $\delta_{i,r}$ are outward residuals for the base tube and coefficients, first and second variations, event quotient and history translation, continuous q/f output action, and full-ball inflation/return domain, then

$$C_i + \sum_r \delta_{i,r} < \mathrm{cap}_i \quad (i = 1, \dots, 6) \quad \Longrightarrow \quad \sup_{z\in B_\lambda} \|B_i(z)\| < \mathrm{cap}_i \quad (i = 1, \dots, 6). \tag{137}$$

Finally, the selected coordinate map $Q_{\mathrm{coord}} : D \to E_s \times \mathbb{R}$ supplied by theorem 4.5 has a qualitative local C^2 stable graph. More precisely, there are neighborhoods U_s of zero in E_s and U of zero in $E_s \times \mathbb{R}$, and a C^2 map $\psi : U_s \to \mathbb{R}$ with $\psi(0) = D\psi(0) = 0$, whose graph consists of the points whose selected-map iterates remain in U and converge to zero. This conclusion has no effective domain radius, graph height, or slope. It does not identify the graph with the periodic orbit's stable-set germ and implies neither a pulse crossing nor biological onset.

Proof. The restriction $Y^* \to E_s^*$ is the metric quotient by $E_s^\perp = \operatorname{span}\{\ell_0, \bar{f}\}$; Hahn–Banach gives equality in equation (135). Apply these exact input costs and the fixed output split term by term in a signed projective representation, then take its infimum, to obtain equation (136). The triangle inequality gives equation (137).

For the last assertion, theorems 4.7 and 4.8 give the hyperbolic linearization $DQ_{\mathrm{coord}}(0) = J^{-1}A^2J$, with stable rate at most 0.01 and unstable inverse rate at most 0.302184. Since Q_{coord} is C^2 on an open neighborhood, the derivative of its nonlinear remainder is arbitrarily small after that neighborhood is shrunk. The local stable- manifold theorem for Banach-space maps gives the asserted C^2 graph, without requiring a self-map of the full anisotropic ball. □

The quotient formula removes phase atoms and multiples of $\bar{f}$ before total variation is taken, while the unstable input is evaluated on $\widehat{q}$ with its sign retained. Thus equation (137) is sharper than multiplying an ambient residual by powers of $\|P_s\|$, but it supplies no numerical ingress by itself. At present the center values and every listed residual are unset: zero of the six numerical Hessian caps in equation (130) has been proved. The local stable graph above therefore cannot be assigned the preferred radius $R_s = 0.0097$, the pulse radius 0.0094, or either quantitative budget in equation (131).

The wide box has the cancellation-blind pair-sum bound 2000, so it violates the older sufficient scalar target $K_{\mathrm{ret}} < 188.912224$ while still closing the correlated split majorant. Hence that scalar target is not a necessary graph-transform input; a source-bound direct return-domain proof may replace it.

There is one indispensable normalization adapter downstream. The graph budgets in equation (131) use the unit- $\widehat{q}$ coordinate, whereas the pulse endpoint intervals use the physical right column. From the proved lower bound $\alpha_q = \|q_{\mathrm{phys}}\|_Y \geq \alpha_{q,-} = 0.07755431589814009$, the conditional physical bounds are

$$\|\psi_{\mathrm{phys}}\| \leq 0.001623186095, \qquad \|D\psi_{\mathrm{phys}}\| \leq 0.3424510530. \tag{138}$$

The height in equation (138) exceeds the older convenient target 0.001; that target was sufficient but not maximal. Direct use of the endpoint-specific intervals gives the strict margins below. Combining these with the proved pulse endpoint and derivative enclosures would give

$$\begin{aligned} &H(J_-) > 0.01805400129, \qquad && H(J_+) < -0.01316048384, \\ &H'(I_J) \subset [-260.79414671, -243.20585329]. \end{aligned} \tag{139}$$

Accordingly, once the proved microscopic event/return tube is enlarged to the declared graph ball, its graph-transform return domain and equation (130) are certified in the identical chart, and once either the direct selected-return hypotheses of theorem 2.6 or an identity $\mathcal{Q} = P^2$ on nested domains is verified, a unique *selected* stable-sheet crossing follows with large strict margin. The center derivative, fixed splitting, intertwinings, and linear rates are already proved. The preferred-domain quantitative stable graph, crossing, stable-set identification, two-sided routing, and biological onset remain unproved. The qualitative selected-map stable graph in theorem 4.11 does not close any of these gates. A no-earlier-hit theorem is needed only for a first-return or ordinal label.

4.4 A correlated wide-parameter pulse family

Let

$$J_0 = \frac{2409}{8000}, \qquad h = \frac{3}{40000}, \qquad J = J_0 + h\zeta, \quad -1 \leq \zeta \leq 1.$$

For the pulse solution $z(t, J)$, propagate the scaled coefficients

$$b_k(t) = \frac{h^k}{k!}\partial_J^k z(t, J_0), \qquad 0 \leq k \leq 4, \tag{140}$$

as one correlated system. Because the vector field is cubic, coefficient products are exact finite convolutions; the degree-five through degree-twelve terms form the forcing for the remainder

$$R_5(t,\zeta) = z(t, J_0 + h\zeta) - \sum_{k=0}^{4} b_k(t)\zeta^k. \tag{141}$$

This scaled representation retains the cancellation that is destroyed by independent derivative boxes.

The 192-bit directed Taylor–Bernstein computation covers all 1152 time cells on $0 \le t \le 24\sqrt{5}$. With time degree 16, every coefficient guide and every remainder tube closes. The certified maxima are

$$\max \|\text{coefficient error}\|_P \le 8.5725360731 \times 10^{-19}, \tag{142}$$

$$\max \|\text{degree 5–12 forcing}\|_P \le 2.0594765514 \times 10^{-9}, \tag{143}$$

$$\max \|R_5\|_P \le 1.7206344728 \times 10^{-8}, \tag{144}$$

$$\max |(R_5)_v| \le 3.5937291129 \times 10^{-9}. \tag{145}$$

The complete computation has been independently regenerated from its fixed inputs. Hence the wide fixed-time parameter family is proved. These bounds do not yet include a moving event time.

4.5 Event alignment and the scalar crossing

Let $g(t,\xi,J)$ define a transverse event and suppose

$$g(\tau(\xi,J),\xi,J) = 0, \qquad |\partial_t g| \ge \sigma > 0. \tag{146}$$

The implicit-function theorem gives

$$\partial_J \tau = -\frac{\partial_J g}{\partial_t g}. \tag{147}$$

Higher event jets follow recursively by differentiating $g(\tau(\xi,J),\xi,J) = 0$. If $z_t(\theta) = z(t+\theta)$, the common-event retained reduced history is

$$\begin{aligned} B_\xi(J) := K_\xi(J) := \Pi_Y z_{\tau(\xi,J)} = \big(\theta \longmapsto v(\tau(\xi,J)+\theta,\xi,J),& \\ w(\tau(\xi,J),\xi,J)\big) \in Y, \qquad -r \le \theta \le 0.& \end{aligned} \tag{148}$$

Thus B_ξ is the notation used in the completion theorem, while K_ξ is retained below for compatibility with the certified event-alignment chain; neither denotes the release history R_ξ. Its two reduced components have derivatives

$$\begin{aligned} \partial_J K_{\xi,v}(J)(\theta) &= v_J(\tau+\theta,\xi,J) + \dot v(\tau+\theta,\xi,J)\partial_J\tau, \qquad -r \le \theta \le 0, \\ \partial_J K_{\xi,w}(J) &= w_J(\tau,\xi,J) + \dot w(\tau,\xi,J)\partial_J\tau. \end{aligned}$$

Thus event alignment is a history translation problem and cannot be replaced by an endpoint correction.

For the physical pulse family in equation (140), let $h_C(\phi) = \phi_v(0) - V_i(0)$, where the exact phase-zero inner-orbit voltage is source-bound by

$$0.9053838432821200 < V_i(0) < 0.9054038432821201. \tag{149}$$

The wider Fourier candidate section used during orbit discovery is not used as a theorem input.

Theorem 4.12 (Uniform Route–C event alignment)**.** *For every $J \in [6021/20000, 753/2500] = [0.30105, 0.30120]$, the exact pulse solution has exactly one positive crossing of $h_C = 0$ in*

$$I_C = \left[\frac{555\sqrt{5}}{24}, 1 + \frac{546\sqrt{5}}{24}\right]. \tag{150}$$

On the whole parameter–time box its event speed satisfies

$$0.2133519018734632 < \dot{v} < 0.2791999723595853. \tag{151}$$

Writing $J = J_0 + h\zeta$, the event time has a fourth-order polynomial $\widehat{T}_4(\zeta)$ such that

$$\left|T(\zeta) - \widehat{T}_4(\zeta)\right| \le 10^{-4}, \qquad -1 \le \zeta \le 1. \tag{152}$$

At J_0,

$$51.79830610110210 < T(0) < 51.79838713426455. \tag{153}$$

The scaled coefficients $\widehat{T}_4(\zeta) = T_0 + \sum_{k=1}^{4} a_k \zeta^k$ satisfy

$$\begin{aligned} &a_1 \in [2.9068862680, 2.9074011687] \times 10^{-2}, \quad && a_2 \in [1.9583818528, 1.9612556829] \times 10^{-4}, \\ &a_3 \in [1.9208606177, 1.9385338925] \times 10^{-6}, \quad && a_4 \in [2.2095125644, 2.2986280783] \times 10^{-8}. \end{aligned} \tag{154}$$

Moreover the translated histories $\mathcal{K}(\zeta) = (v(T(\zeta) + \cdot, J), w(T(\zeta), J)) \in Y$ obey

$$\|\mathcal{K}(\zeta) - B_c(\zeta)\|_Y \le 0.008199931248423, \tag{155}$$

where the continuous fourth-order fixed-time reference family is

$$B_c(\zeta) = \left(\theta \longmapsto \sum_{k=0}^{4} b_{k,v}(T_0 + \theta)\zeta^k, \sum_{k=0}^{4} b_{k,w}(T_0)\zeta^k\right).$$

The proof combines the source-bound fixed-time family with cellwise Bernstein endpoint and vector-field bounds. The gap is uniformly negative at $\widehat{T}_4 - 10^{-4}$, with upper bound -6.4436×10^{-7}, and uniformly positive at $\widehat{T}_4 + 10^{-4}$, with lower bound 6.4342×10^{-7}. The positive speed then gives existence and uniqueness. This proves neither that the event is the third post-release crossing nor that $\mathcal{K}(\zeta)$ lies on the inner stable sheet. A fourth-order Y-valued history jet is not claimed. The first derivative, including the moving-event contribution, is certified next.

Theorem 4.13 (Event-aligned first pulse derivative)**.** *Put $K(J) = \mathcal{K}((J - J_0)/h)$. The map*

$$K : [0.30105, 0.30120] \longrightarrow Y$$

is continuously differentiable. If $W = \partial_\zeta z$, $C = \partial_\zeta \sum_{k=0}^{4} b_k \zeta^k$, and $E = W - C$, the directed comparison for E closes on all 1152 cells with maximum weighted radius

$$8.7288930458 \times 10^{-8}. \tag{156}$$

The event derivative satisfies

$$T_J \in [336.6243028349, 456.5740938122]. \tag{157}$$

On the complete event history,

$$\begin{aligned} D_J K_v(0) &= 0, \\ D_J K_w &\in [-17.350656459, -10.142798861], \\ \|D_J K\|_Y &\leq 142.200202974. \end{aligned} \tag{158}$$

For the fixed center Route–C functional from equations (103) *and* (105)*, the presently directed consequence is only*

$$|f(D_J K)| \leq 1017.273043. \tag{159}$$

No sign or exclusion of zero is asserted in equation (159)*.*

The proof does not differentiate the Stage-5B remainder bound. The exact error equation is

$$E' = DF(z)E + \{DF(z) - DF(B)\}C + \partial_\zeta \operatorname{Tail}_{\geq 5}(B), \qquad B = \sum_{k=0}^{4} b_k \zeta^k.$$

The state remainder enters only through the correlated linearization difference. A 192-bit Taylor–Bernstein comparison on 64 parameter shards closes every fixed-time cell. The event identity

$$T_\zeta = -W_v(T,\zeta)/\dot{v}(T,\zeta)$$

is then evaluated on 128 parameter shards, and the history chain rule retains $z_t T_J$ on the entire translated window. Finally, equation (159) uses the Stage-4D density total variation and atom moduli. This last operation deliberately loses orientation.

Lemma 4.14 (Phase-invariant oriented quotient)**.** *Let ℓ be a nonzero complex representative of the left eigenspace of the simple real unstable multiplier, and let $q, y \in Y$ be real with $\ell(q) \neq 0$. Suppose one common directed computation gives*

$$|\ell(y) - a_0| \leq r_a, \qquad |\ell(q) - b_0| \leq r_b < |b_0|\,.$$

With $c_0 = a_0/b_0$, the normalized action is real and obeys

$$\frac{\ell(y)}{\ell(q)} \in [\Re c_0 - r_c, \Re c_0 + r_c], \qquad r_c = \frac{r_a + |c_0|\, r_b}{|b_0| - r_b}. \tag{160}$$

Equivalently, for any real c, if the same row gives $|\ell(y - cq)| \leq r$ and $|\ell(q)| \geq b_- > 0$, then

$$\frac{\ell(y)}{\ell(q)} \in [c - r/b_-, c + r/b_-]. \tag{161}$$

Indeed the quotient is invariant under every nonzero complex rescaling of ℓ. Reality and simplicity imply that conjugation changes ℓ only by such a rescaling, so its action ratio on real histories is real. The disk estimate in equation (160) follows by writing the difference from a_0/b_0 over the common denominator. Equation (161) follows from

$$\frac{\ell(y)}{\ell(q)} - c = \frac{\ell(y - cq)}{\ell(q)}.$$

Equation (161) is the preferred Stage-5E form: choose a source-bound center c, form $D_J K - cq$ as one complete-history object, apply the same Fourier row and tail enclosure, and divide only after the correlated residual action has been bounded. This is the first-variation analogue of the successful same-row deflation in theorem 4.1.

Lemma 4.15 (Physical right-gauge correction)**.** *Let $\widetilde{q}$ be a complex Grushin-normalized right eigencolumn for a simple real multiplier of a real return operator. Choose a real evaluation χ with $\chi(\widetilde{q}) \neq 0$, and orient the physical unstable history by*

$$\gamma = \frac{\chi(\widetilde{q})}{|\chi(\widetilde{q})|} \in \mathbb{S}^1, \qquad q = \gamma^{-1}\widetilde{q}.$$

Then q is real and $\chi(q) > 0$. For any raw left representative ℓ with $\ell(\widetilde{q}) \neq 0$, the real normalized unstable functional and stable projection on real histories are

$$f(y) = \gamma\frac{\ell(y)}{\ell(\widetilde{q})}, \qquad qf(y) = \widetilde{q}\frac{\ell(y)}{\ell(\widetilde{q})}. \tag{162}$$

In particular, the second expression is independent of both complex left- and right-eigenvector phase gauges.

To prove the lemma, reality and simplicity imply $\widetilde{q} = \eta q_{\mathbb{R}}$ for a real eigenhistory $q_{\mathbb{R}}$ and some $\eta \neq 0$. Orienting $q_{\mathbb{R}}$ by $\chi(q_{\mathbb{R}}) > 0$ shows that $\gamma = \eta/|\eta|$, so $\gamma^{-1}\widetilde{q}$ is real. The identities in equation (162) then follow by normalization.

Theorem 4.15 is not optional in the executable certificate. The stored Grushin eigencolumn carries a nontrivial complex phase even though the exact multiplier is real. Dividing $\ell(D_J K)$ directly by $\ell(\widetilde{q})$ therefore produces a complex, right-gauge-dependent number; it is not the oriented action in theorem 4.14. The Stage-5E residual must first use the physical real history $q = \gamma^{-1}\widetilde{q}$, or equivalently insert the factor γ in equation (162). This correction moves the diagnostic center from the recovery-atom shard near -14 to the normalized physical value near -252.

Theorem 4.16 (Fixed-center physical-phase action of the event derivative)**.** *Let q_{phys} and f_{phys} be the fixed center Grushin pair oriented by theorem 4.15, with $f_{\mathrm{phys}}(q_{\mathrm{phys}}) = 1$. For every $J \in [0.30105, 0.30120]$,*

$$f_{\mathrm{phys}}(D_J K(J)) \in [-258.746521015805, -245.253478984195] \subset (-\infty, 0). \tag{163}$$

The enclosure is uniform on the full pulse interval and contains the event-time translation term. It is an action of the fixed normalized Route–C functional, not yet a stable-sheet-gap derivative.

The source-bound calculation takes $c_* = -252$, forms the complete history $Y_*(J) = D_J K(J) - c_* q_{\mathrm{phys}}$, and keeps the recovery atom and both delayed-density pieces in one directed action. The current voltage atom vanishes exactly because $D_J K_v(0) = (q_{\mathrm{phys}})_v(0) = 0$. On 128 pulse-parameter shards and 512 exact history cells, the two additive joint acceptance envelopes give

$$\begin{aligned} r_{\mathrm{guide}} &\leq 0.002869679658061285, \\ r_{\mathrm{measure}} &\leq 1.899608044833 \times 10^{-6}, \\ r_A &\leq 0.002871579266106117. \end{aligned} \tag{164}$$

The same-row denominator satisfies

$$|\ell(q_{\mathrm{phys}})| \geq 0.0004256385267872229. \tag{165}$$

Thus equation (161) gives radius at most 6.746521015805 about c_*, proving equation (163). All arithmetic is outward at 192 bits; the finite row, Neumann tail, orbit covariance, density-basis transfer, convolution rounding, pulse remainder, event graph, and seams are covered before the final division.

The executable biological-control contract imports exactly six outputs from theorem 4.12: the section level, pulse interval, unique declared- bracket event, speed interval, time-graph remainder, and common-event Y-tube. Its fields for J_c, stable-gap signs, physical onset, routing, and capture remain null or false; its optional interval-Newton enclosure is likewise absent. Event alignment therefore becomes a proved input to the control chain without silently promoting its next implication.

Once the graph ψ_ξ and normalized functional f_ξ are available, let $\phi^{\Sigma}_{i,\xi}$ be the inner-orbit base history on the phase section and define the signed stable-sheet gap

$$H(\xi, J) = f_\xi(K_\xi(J) - \phi^{\Sigma}_{i,\xi}) - \psi_\xi\big[(\mathrm{I} - q_\xi f_\xi)(K_\xi(J) - \phi^{\Sigma}_{i,\xi})\big]\,. \tag{166}$$

For each fixed ξ, the pair (q_ξ, f_ξ) in this formula is held fixed as J varies. If $[F_J]$ encloses the oriented quotient $f_\xi(D_J K_\xi)$, $\|D\psi_\xi\| \le L_\psi$, and $\|(\mathrm{I} - q_\xi f_\xi)D_J K_\xi\| \le M_s$, then the exact correlated derivative satisfies

$$\partial_J H(\xi, J) \in [F_J] + [-L_\psi M_s, L_\psi M_s]. \tag{167}$$

The oriented action and graph correction enter this estimate separately; the norm in equation (158) alone gives no slope.

At the center parameter, fix the exact reduced history space and Route–C section

$$Y = C([-\tau_{\max}, 0], \mathbb{R}) \times \mathbb{R}, \qquad \Sigma = \{y \in Y : y_v(0) = 0\},$$

with the inherited max norm. Let q_{phys} and f_{phys} be the same fixed physically oriented Grushin pair used in theorem 4.16, put $P_s = \mathrm{I} - q_{\mathrm{phys}} f_{\mathrm{phys}}$, and set $\kappa(J) = K(J) - X_*$, where X_* is the inner-orbit reference history on this section. Thus the center specialization of equation (166) is $H(J) = f_{\mathrm{phys}}(\kappa(J)) - \psi(P_s\kappa(J))$.

Lemma 4.17 (Global affine splitting of the Route–C section)**.** *Let $q \in \Sigma$ and $f \in \Sigma^*$ satisfy $f(q) = 1$, and put $P_s = \mathrm{I} - qf$. Then* $\operatorname{ran} P_s = \ker f$, *and*

$$\mathcal{C} \colon \Sigma \longrightarrow \ker f \times \mathbb{R}, \qquad \mathcal{C}(y) = (P_s y, f(y)) \tag{168}$$

is a bounded linear isomorphism with inverse $(z, u) \mapsto z + qu$. Hence subtraction of X_* gives global coordinates on the affine section $X_* + \Sigma$. If $\psi \colon D_s \subset \ker f \to \mathbb{R}$ represents the local stable sheet as

$$X_* + \{z + q\psi(z) : z \in D_s\}, \tag{169}$$

then, for every section history K with $P_s(K - X_*) \in D_s$,

$$K \text{ belongs to this graph} \quad \Longleftrightarrow \quad f(K - X_*) - \psi(P_s(K - X_*)) = 0. \tag{170}$$

In particular, no additional numerical containment condition for an ambient Route–C chart is required; only containment of the stable coordinate in D_s is needed.

Indeed, $f(P_s y) = 0$ and $y = P_s y + qf(y)$. These identities prove both the range statement and the two inverse relations in equation (168); equation (170) then follows from uniqueness of the split coordinates. This global affine coordinate statement does not enlarge the local domain D_s on which the stable graph has been constructed.

Theorem 4.18 (Centered endpoint coordinates, stable cone, and conditional slope)**.** *On $I_J = [J_-, J_+] = [0.30105, 0.30120]$, source-bound complete-history endpoint calculations give*

$$\begin{aligned} &f_{\mathrm{phys}}(\kappa(J_-)) \in F_- := [0.019677187, 0.023055896], \\ &f_{\mathrm{phys}}(\kappa(J_+)) \in F_+ := [-0.018164491, -0.014783669], \\ &\|P_s\kappa(J_-)\|_Y \le 0.008935972, \qquad \|P_s\kappa(J_+)\|_Y \le 0.008927665. \end{aligned} \tag{171}$$

The direct norm of the same physical unstable column and the correlated projection identity give

$$\|q_{\mathrm{phys}}\|_Y \leq 0.086274581, \qquad \sup_{J\in I_J} \|P_s D_J \kappa(J)\|_Y \leq 5.979324. \tag{172}$$

A two-ended Banach-space cone, using both bounds in equation (171)*, then proves*

$$P_s\kappa(I_J) \subset \overline{B}_Y\left(0, \frac{47}{5000}\right) \cap \ker f_{\mathrm{phys}}. \tag{173}$$

Theorem 4.3 supplies the discrete stable-power input $K_s = 1, \rho_s = 0.1$ for a future graph transform. It supplies neither the nonlinear return tube nor any of the six uniform Hessian blocks. If a quantitative centered C^1 graph $\psi\colon D_s \subset \ker f_{\mathrm{phys}} \to \mathbb{R}$ is constructed using this same $Y, \Sigma, q_{\mathrm{phys}}, f_{\mathrm{phys}}, P_s$, if

$$\overline{B}_Y\left(0, \frac{47}{5000}\right) \cap \ker f_{\mathrm{phys}} \subset D_s, \qquad \sup_{J\in I_J} \|D\psi(P_s\kappa(J))\| \leq 16, \tag{174}$$

then

$$H'(J) \in [-354.415700, -149.584300] \subset (-\infty, 0) \qquad (J \in I_J). \tag{175}$$

The corresponding maximum admissible graph-derivative norm in this budget is strictly larger than 41.016926*.*

For completeness, put $z(J) = P_s\kappa(J)$ and $W = J_+ - J_- = 3/20000$. From equation (172), the fundamental theorem of calculus gives

$$\|z(J)\|_Y \leq \min\{E_- + 5.979324(J - J_-), E_+ + 5.979324(J_+ - J)\},$$

where $E_- = 0.008935972$ and $E_+ = 0.008927665$ are the directed endpoint bounds above. The exact-rational intersection of these two affine cones lies strictly inside I_J, and its directed maximum is at most $0.009380268 < 47/5000$, proving equation (173). Combining equations (167) and (172) with equation (163) proves the conditional interval equation (175). Thus the endpoint functional signs, the full-interval stable-coordinate ball, and this conditional slope arithmetic are proved. The theorem does not supply the graph, either condition in equation (174), endpoint graph heights, graph-adjusted stable-gap signs, a selected crossing, onset, routing, capture, or network safety.

Proposition 4.19 (Completion of the selected center-parameter crossing)**.** *Let $I_J = [J_-, J_+] = [0.30105, 0.30120]$, and let $K \in C^1(I_J, X_* + \Sigma)$ be the single selected Route–C event branch enclosed in theorems 4.12 and 4.13. Suppose a quantitative centered C^1 stable graph $\psi\colon D_s \subset \ker f_{\mathrm{phys}} \to \mathbb{R}$ is constructed in the identical $Y, \Sigma, q_{\mathrm{phys}}, f_{\mathrm{phys}}, P_s$ normalization and*

$$\overline{B}_Y\left(0, \frac{47}{5000}\right) \cap \ker f_{\mathrm{phys}} \subset D_s, \qquad \sup_{J\in I_J} \|D\psi(P_s\kappa(J))\| \leq 16. \tag{176}$$

Let F_-, F_+ be the proved functional intervals in equation (171)*. Assume graph-height bounds $\eta_-, \eta_+ \geq 0$ such that*

$$\begin{aligned} |\psi(P_s\kappa(J_-))| &\leq \eta_-, \\ |\psi(P_s\kappa(J_+))| &\leq \eta_+, \end{aligned} \tag{177}$$

and that the graph-adjusted margins are strict:

$$\inf F_- - \eta_- > 0, \qquad \sup F_+ + \eta_+ < 0. \tag{178}$$

Then there is a unique

$$J_c^{\mathrm{sel}} \in (J_-, J_+) \quad \textit{such that} \quad H(J_c^{\mathrm{sel}}) = 0, \tag{179}$$

and $K(J_c^{\mathrm{sel}})$ lies on the local stable sheet in equation (169)*. Moreover the intersection is transverse, with $H'(J_c^{\mathrm{sel}}) \leq -149.584300 < 0$.*

To prove the proposition, equations (171), (177) and (178) give $H(J_-) > 0 > H(J_+)$. The gap is continuous, so the intermediate value theorem gives a zero. Conditions equation (176) and theorem 4.18 make H strictly decreasing on all of I_J, proving uniqueness and transversality; the stable-sheet statement follows from the affine splitting lemma above. Interval Newton may subsequently replace I_J by a smaller certified enclosure of J_c^{sel}, but it is not needed for either existence or uniqueness.

The conclusion of the proposition is one scalar at the center parameter $\xi_0 = (1/4, 1/200)$; it does not define a C^1 function $J_c(\xi)$. Such a function requires a common parameter neighborhood on which $K_\xi(J), X_{*,\xi}, q_\xi, f_\xi, P_{s,\xi}$, and ψ_ξ are jointly C^1 in compatible stable-bundle coordinates, together with uniform stable-coordinate domain containment, endpoint margins, and a bound $\partial_J H(\xi, J) \le -m_J < 0$. Under those additional hypotheses the implicit-function theorem gives

$$D_\xi J_c = -\frac{D_\xi H}{\partial_J H}. \tag{180}$$

None of these parameter-uniform graph and crossing hypotheses is supplied by the center-parameter proposition. Even that proposition proves a selected stable-sheet intersection, not a biological onset. To obtain the latter, invariant exit cones must carry the two local sides to compact exit faces, and finite-time enclosures must attach those faces to the quiet and outer attraction tubes. The strict quiet capture at $J = 0.30$ supplies one anchor; the corresponding outer-side tube is not yet complete.

4.6 Frequency–amplitude–threshold target coordinates

Conditional on the parameter-uniform joint-C^1, endpoint-margin, and strict-slope hypotheses above, let $P(\xi) = (F(\xi), A(\xi))$ and define the selected-threshold offset and target map

$$S(\xi, J) = J - J_c(\xi), \qquad \mathcal{Q}(\xi, J) = (P(\xi), S(\xi, J)).$$

This is a local target coordinate relative to the selected stable sheet. It is not yet a biological safety coordinate: that interpretation requires the additional uniform routing conclusion. Algebraically,

$$D\mathcal{Q} = \begin{pmatrix} B & 0 \\ -c^T & 1 \end{pmatrix}, \qquad B = D_\xi P, \quad c = D_\xi J_c, \tag{181}$$

so $\det D\mathcal{Q} = \det B$. The outer periodic branch has the directed bounds

$$\det B \in [0.533847, 1.694625], \qquad \left\|B^{-1}\right\|_2 \le 22.044336699647401, \tag{182}$$

and its branch-centered (F, A) target-ball radius is at least $4.5363124943 \times 10^{-12}$.

The determinant identity does not quantify the three-output inverse. If $\left\|B^{-1}\right\| \le K$ and $\|c\| \le L_c$, then

$$(D\mathcal{Q})^{-1} = \begin{pmatrix} B^{-1} & 0 \\ c^T B^{-1} & 1 \end{pmatrix}.$$

Consequently

$$\left\|(D\mathcal{Q})^{-1}\right\|_2^2 \le \Lambda_+(K, L_c), \tag{183}$$

where

$$\Lambda_+(K, L) = \frac{1}{2}\left[K^2(1 + L^2) + 1 + \sqrt{\{K^2(1 + L^2) - 1\}^2 + 4L^2K^2}\right].$$

The selected-threshold slope affects conditioning, but not rank.

There is also an exact nonlinear product formulation. If P maps a parameter neighborhood U bijectively onto V, then for every $\sigma > 0$

$$\mathcal{Q} : \{(\xi, J) : \xi \in U,\ |J - J_c(\xi) - S_0| < \sigma\} \longrightarrow V \times (S_0 - \sigma, S_0 + \sigma) \tag{184}$$

is bijective. For a rectangular actuator neighborhood $\|\xi - \xi_0\| \le R_\xi$, $|J - J_0| \le R_J$, a centered Euclidean output ball of radius ρ is contained whenever

$$\rho \le \rho_{FA}, \qquad K\rho \le R_\xi, \qquad (L_c K + 1)\rho \le R_J. \tag{185}$$

One must also keep $J_0 \pm (L_c K + 1)\rho$ in the validated physical pulse interval. These inequalities make the missing role of equation (180) explicit.

4.7 Network-robust local threshold erosion

For the left-balanced subclass $\delta_B = 0$, let R_0 be the maximum of the collective entrance error and the retained transverse-history diameter. Suppose a source-bound gap-response estimate provides constants $L_{H0}, L_{H1}, L_{HJ0}, L_{HJ1}$ such that

$$\begin{aligned} \left\|\widetilde{H} - H\right\|_\infty &\le \epsilon_H(R_0) := L_{H0}R_0 + L_{H1}\frac{56483}{3200}R_0^2, \\ \left\|\partial_J \widetilde{H} - \partial_J H\right\|_\infty &\le \delta_{HJ}(R_0) := L_{HJ0}R_0 + L_{HJ1}\frac{56483}{3200}R_0^2. \end{aligned} \tag{186}$$

If the oriented scalar slope satisfies $-\partial_J H \ge m_J > 0$ on $[J_c - r_J, J_c + r_J]$ and

$$\epsilon_H(R_0) < m_J r_J, \qquad \delta_{HJ}(R_0) < m_J, \tag{187}$$

then the network gap has opposite endpoint signs and negative derivative. This is the source-bound increasing-gap perturbation lemma applied to $-H$; no change of constants is involved. The monotone-root perturbation lemma gives exactly one $J_{c,N}$ in $[J_c - r_J, J_c + r_J]$ and

$$|J_{c,N} - J_c| \le \Delta_J(R_0) := \frac{\epsilon_H(R_0)}{m_J}. \tag{188}$$

No root outside that local interval is excluded. If the pulse actuator has amplitude error at most e_J, write $J_{\mathrm{cmd}} = J_c + S_{\mathrm{cmd}}$, assume $|J_{\mathrm{real}} - J_{\mathrm{cmd}}| \le e_J$, and require every realized amplitude under consideration to remain in $[J_c - r_J, J_c + r_J]$. The realized pulse then lies on the side of the local network root selected by S_{cmd}, uniformly in topology, whenever

$$|S_{\mathrm{cmd}}| > \Delta_J(R_0) + e_J. \tag{189}$$

Because the network gap is decreasing, its sign at the realized pulse is opposite to the sign of S_{cmd}. For a target ball centered at S_0 with threshold-offset radius ρ_S, the whole ball stays on one local network-gap side provided

$$|S_0| - \rho_S > \Delta_J(R_0) + e_J.$$

This is the precise local selected-root erosion law for the abstract balanced finite-network composition under equations (186) and (187). It is independent of finite network size and admitted topology whenever the gap-response constants are common. If, in addition, a common product lift and two-sided routing theorem identify the two local gap sides with the quiet and outer basins, then the same inequality becomes a routed biological safety guard. The upper-triangular theorem also bounds nonbalanced collective forcing by equation (97), but no nonbalanced routing or gap-response certificate has been supplied. At present the four response constants, m_J, and the scalar/product routing constants are not all numerically validated, so no concrete asynchronous biological safety radius is claimed in either class.

5 Claim ledger, reproducibility, and outlook

5.1 Four threshold and coordinate notions

The terminology of the paper is fixed by the following distinctions.

(a) A *selected canard root* is a zero of a normalized complete-history Lin or graph–trace gap. It depends on the declared phase and preparation until a zero-fiber theorem identifies the selected histories geometrically.

(b) A *selected pulse/stable-sheet threshold* is a zero of equation (166). It is defined by an actual physical pulse history, the declared event branch and section, and the local stable manifold of the inner periodic orbit.

(c) A *biological onset* is a selected pulse/stable-sheet threshold together with two-sided routing to the quiet and outer attracting objects.

(d) The signed offset $S = J - J_c(\xi)$ is always a *selected-threshold target coordinate* once J_c has been proved. It is a *biological safety coordinate* only on a neighborhood where the preceding onset and routing are uniform.

There is no implication between (a) and (b) without a separate comparison theorem; (c) and the safety interpretation in (d) require the additional routing conclusion stated explicitly above. In particular, a projected voltage event is neither a complete-history canard root nor a basin separator.

5.2 Evidence ledger

Ingredient	Status	Evidence boundary
Finite-network transverse decay and complete-line inverse	Proved	Every finite nonnegative row-mass Dobrushin topology in equation (5); rate 1/10, inverse bound 10, fixed voltage strip. No common delay-layer left balance is required.
Canonical synchronized root transfer	Proved implication	Exact upper-triangular factorization and the bounded invertible codomain row reduction equation (54); the full cokernel is equation (55), with $\|C_N G_{\perp,N}\| \le 391\delta_B/2000$. The scalar phase-fixed complete-history root is a hypothesis, not supplied here.
Dimension-uniform one-gap theorem	Proved implication	Explicit radius and quadratic remainder under uniform range, trace, fitting, regularity, and simple-root bounds.
Shared-resource heterogeneous-curvature instance	Proved	Arbitrary finite size, common Dobrushin gap, fixed-support row-neutral direction, no nontrivial synchrony polydiagonal.
Eventually smooth selected-event return	Proved, general RFDE theorem	For $X = C([-\tau_*, 0], \mathbb{R}^d)$, finite d, a C^k fixed-delay functional and initial-section chart, a common solution/event tube, an open C^k section-functional domain, and a selected window beginning after $k\tau_*$, strict endpoint signs and positive speed imply a unique C^k event-hit map and, when it returns to the same chart, a C^k section map. No no-earlier-hit premise is needed; the sufficient threshold is independent of finite network size.

Ingredient	Status	Evidence boundary
Direct selected-return stable-set identification	Proved, general semiflow lemma	A bounded positive-time selected return of the same semiflow, with a common intervening tube, a phase-isolating local section, invariant return domain, and recurrent selected hits on stable flow orbits, identifies the intrinsic periodic-orbit stable-set germ. An identity $Q = P^m$ and a first-return label are sufficient but unnecessary.
Nonlinear synchronization and collective defect	Proved in strip	$\exp(-t/10)$ diameter decay. At $\delta_B = 0$, the pointwise history-envelope constant is 703/200 and the accumulated constant is $703/40 + 27\sqrt{5}/800 < 56483/3200$. Without left balance, equations (95) and (97) add the resolved linear imbalance. No strip, scalar route, product lift, basin, or onset follows.
Inner one-unstable splitting, spectral gap, and adjoint	Proved	One nontranslation unstable exponent, stable radius at most $e^{-0.01}$, continuous RFDE adjoint measure and nonzero normalization.
Base-orbit correlated uu block	Proved	Physical-time continuous-history second variation and same-row deflation; $C^{uu}_{s,\mathrm{base}} < 7.905650 < 12$ in equation (108). Not uniform on the split ball.
Selected inner terminal stable row	Proved, discrete linear	For $A = \Pi_T U(T, 0)\|_{\Sigma_0}$ and $P_s = \mathrm{I} - qf$, $\|AP_s\|_{\Sigma_0 \to Y} \leq 0.009896427481610001 < 0.1$. Exact intertwining gives $\|A_s^n\|_{E_s \to Y} \leq 0.1^n$ with $K_s = 1$. This is the selected phase-fixed map only, not a first-return or nonlinear-tube result.
Moving-event return Hessian	Proved analytic identity	Theorem 4.4 includes both event-time derivatives, translation of every complete-history coordinate, and exactly one moving phase projection before the fixed stable/unstable output split. It explains why the current one-period argument does not establish C^2 on an arbitrary continuous-history ball; no non-C^2 theorem is claimed. It supplies no model-specific full-ball event tube or Hessian cap.
Near-two-period smoothing and signed Hessian pilot	Center gate proved; pilot nonrigorous	The exact center clears the two-delay C^2 threshold by more than 14.010740. The source-bound $n_{\mathrm{step}} = 120, 180, 240$ signed finite-section pilot gives the heuristic envelope $(2.36 \times 10^{-6}, 0.00873, 32.118, 0.597, 0.578, 158.532)$, entirely inside the simultaneous wide box. This is not a continuous-history error bound or a Hessian certificate.
Explicit common near-two-period event tube	Proved, microscopic scale	On the full arbitrary-continuous-history ball $\lambda_0 = 9 \times 10^{-31}$, one common physical window has strict endpoint signs, event speed greater than 0.244880, a unique selected event, terminal local-section containment, and a C^2 complete-history hit. The proof uses a strictly larger open scale 9.1×10^{-31}. It is not a same-ball self-map, a preferred-scale result, or an event-ordinal theorem.

Ingredient	Status	Evidence boundary
Orbit-aware enlarged near-two-period event tube	Proved, explicit scaled ball	Theorem 4.6 uses the exact weighted-energy cancellation and a 1042-cell, 192-bit directed orbit integral to prove the same selected C^2 hit on the closed scale 1.2×10^{-8}, with a strictly larger open scale 1.25×10^{-8} and sufficient-construction ceiling greater than $1.2770076307683837 \times 10^{-8}$. This is about 22 decimal orders larger than the scalar comparison, but it is not a preferred-scale self-map, Hessian, graph, crossing, or onset theorem.
Direct two-period derivative bridge	Proved	For the direct reduced physical hit map, $DQ_Y(Y_*) = A^2$ by the periodic variational cocycle and exact removal of the intermediate phase-tangent component. In section coordinates the derivative is $J^{-1}A^2J$; in the compatible full history space it is intertwined by the lift. No nonlinear identity $Q = P^2$ is used.
Fixed two-step linear splitting and rates	Proved, linear	For $B = A^2$, the same complementary $E_s \oplus E_u$ splitting and exact intertwinings hold. The model instance has $K_{s,2} = K_{u,2} = 1$, $\rho_{s,2} \le 0.01$, and unstable inverse/backward rate $\rho_{u,2} \le 0.302183501335054$; the forward unstable multiplier still has modulus greater than one. A nonlinear identity $Q = P^2$ requires the nested domain $D \cap P^{-1}(D)$, but is not necessary for a direct selected return.
Finite-sampling Hessian obstruction and atomic lift	Proved	On the arbitrary C^0-history unit ball, $\|\mathrm{I} - R_N S_N\| \ge 1$ for every finite sampler and reconstruction. Therefore the Stage–4Q mesh ladder cannot yield a Banach Hessian error merely by refinement. A rigorous upgrade must enclose a directed signed atom–density/bimeasure residual, event quotient, exact continuous q/f action, output-phase completion, and full-ball inflation. The atomic lift bound $\|B_N\| \le \max_o \sum_{ij} \|a_{oij}\|$ is proved, but no one of the six actual block bounds is promoted.
Exact split quotient, residual implication, and local selected graph	Proved qualitative and analytic statements	Theorem 4.11 proves the exact stable-row quotient norm modulo $\mathrm{span}\{\ell_0, \bar{f}\}$, the corresponding split-projective signed-bimeasure bound, and the strict center-plus-residual implication for all six Hessian blocks. It also proves a local C^2 stable graph for the selected coordinate map on an unspecified neighborhood. Numerical ingress is 0/6: there is no effective radius, graph height, or slope, no preferred-scale graph, and no identification with the periodic orbit's stable-set germ. Thus no pulse crossing or onset follows.

Ingredient	Status	Evidence boundary
Two-return graph-closure arithmetic	Proved conditional implication	Under a preferred-ball extension/return domain and the six caps $(1, 10, 1000, 5, 10, 1000)$, the split majorant has Perron bound below 0.193086, unit-coordinate height below 1.25886×10^{-4}, and derivative below 0.026559. After the proved physical normalization, the conditional height is below 0.001623187, the endpoint margins exceed 0.0180540 and 0.0131604, and $H' \subset [-260.794147, -243.205853]$. The center derivative, fixed splitting, and rates are proved; the preferred-domain and Hessian premises are not.
Quantitative inner stable graph	Open	A qualitative local C^2 stable graph for the selected coordinate map is proved. What remains open is an effective graph on the declared anisotropic pulse scale: graph-transform return-domain containment and all six directed continuous-history Hessian bounds in equation (130). Identifying that selected-map graph with the periodic orbit's stable-set germ then requires either the direct-return hypotheses of theorem 2.6 or $Q = P^2$ on nested domains. No-earlier-hit is needed only for a first-return or ordinal label, not for the selected crossing or routing completion theorem.
Wide physical-pulse parameter family	Proved at fixed time	All 1152 cells and the full interval $J \in [0.30105, 0.30120]$; fourth-order correlated jet and fifth-order remainder equation (144).
Uniform Route–C event alignment	Proved	Exactly one positive event in equation (150) for every J in the full interval; speed equation (151), fourth-order time graph equation (152), and continuous Y-history tube equation (155). No ordinal or stable-sheet claim.
Event-aligned first pulse derivative	Proved	Direct first-variation enclosure on all 1152 cells; T_J in equation (157), the complete-history translation term, and $\|D_J K\|_Y$ in equation (158). The fixed-functional result equation (159) is only a modulus bound containing zero.
Physical-phase oriented functional action	Proved	The physical right gauge and same-row residual give $f_{\mathrm{phys}}(D_J K(J)) \in [-258.746522, -245.253478]$ on the full pulse interval; see theorem 4.16. This is not a stable-gap slope.
Endpoint functional coordinates and full stable-coordinate cone	Proved	Source-bound complete-history endpoint calculations give $f_{\mathrm{phys}}(\kappa(J_-)) > 0 > f_{\mathrm{phys}}(\kappa(J_+))$, with $\|P_s\kappa(J_-)\|_Y \le 0.008935972$ and $\|P_s\kappa(J_+)\|_Y \le 0.008927665$. The direct physical-column norm and a two-ended Banach-space cone prove $P_s\kappa(I_J) \subset \overline{B}_Y(0, 47/5000) \cap \ker f_{\mathrm{phys}}$. These are functional signs and stable-coordinate containment, not graph-adjusted stable-gap signs or graph-domain containment.

Ingredient	Status	Evidence boundary
Centered stable-gap slope bridge	Proved conditional implication	In the identical Y/Σ norm, physically oriented Grushin pair, and centered coordinate $\kappa = K - X_*$, $\|P_s D_J \kappa\|_Y \leq 5.979324$. A future graph whose stable domain contains the certified radius-47/5000 ball and whose derivative norm is at most 16 would satisfy $H' \subset [-354.415700, -149.584300]$. The sharper two-return wide-box adapter would give $[-260.794147, -243.205853]$, but only after its full-ball graph premises are proved. Both implications are rigorous arithmetic; the graph remains open.
Selected pulse/stable-sheet threshold	Open	The event history, first derivative, endpoint functional signs, radius-47/5000 stable-coordinate cone, and conditional centered slope bridge are enclosed. The remaining release chain is the source-bound two-return map, its fixed splitting/rates and six-block graph theorem on that ball, followed by direct-return stable-set identification or a nested- domain $Q = P^2$ identity. The proved wide-box adapter then supplies graph-adjusted endpoint signs and strict monotonicity, hence one unique selected crossing; interval Newton is only an optional sharpening.
Quiet-side capture	Proved anchor	The complete history generated at $J = 0.30$ lies strictly in an explicit quiet Razumikhin basin.
Outer resolvent and center-guide combined rows	Proved v2	Stage 3G covers all $730 + 40$ rectangles and 12,320 patches and proves the directed tensor residual and Green/boundary bootstrap. Stage 3H proves strict center-guide phase-combined full-row bounds 41.482447 in voltage and 1.177738 in recovery. The incomplete v1 values remain withdrawn.
Outer continuous-density transfer and C^0 contraction	Proved linear	Stage 3I covers all 25,600 event-aware density rectangles and proves exact reduced-history row bounds 0.550516 in voltage and 0.028281 in recovery. Hence the phase-fixed linear return contracts arbitrary reduced C^0 section histories. The auxiliary 0.01 closeness target fails but is not the contraction gate.
Nonlinear outer local return and attracting tube	Proved locally	Stage 6A proves a C^2 next-event map with no earlier local-section hit, a nonlinear return contraction on the radius-10^{-335} reduced section ball, and a compatible invariant complete-history return tube of radius 10^{-3}. Its return constant satisfies $\Lambda_6 \leq 0.561839$. This is a microscopic local attraction theorem, not pulse capture.

Ingredient	Status	Evidence boundary
$J = 0.32$ entry, outer-side attachment, and capture	Open	The ambient history lies within 2.637079×10^{-5} of the phase-zero outer history, but it has not been placed on the exact section and lies far outside the Stage-6A domain even under optimistic comparison. The first failed gate is ambient-to-section domain containment. At that attachment scale the nonlinear return budget requires $C_P < 17044.784546 < 17044.785$. Outer-side attachment, capture, routing, and onset remain open.
Outer frequency–amplitude response	Proved	Directed determinant interval, inverse bound, and nonzero branch-centered target ball equation (182).
Three-output threshold target map	Conditional	The block inverse and local root-erosion formulas are proved implications; their model-specific J_c, $D_\xi J_c$, and asynchronous response constants are not yet all validated. The third output becomes a biological safety coordinate only after uniform two-sided routing is proved.

5.3 Source-bound validation

Every numerical statement promoted above is represented by four linked objects: a proof-oriented source module, a deterministic generator, a serialized result, and hostile regression tests. The serialized manifest records hashes of its mathematical sources and parents. Validation recomputes the certificate or verifies an independently regenerated copy, checks the parent hashes, and rejects mutation of an open flag to a proved flag. Mesh refinement is never interpreted as interval error. For Stage 4L the frozen source manifest additionally binds the design contract, theorem note, hostile test module, and directly used MPFR directed-interval implementation.

The principal releases integrated in this revision are frozen at the following raw-result and canonical certificate, contract, design, pilot, bridge, or artifact SHA-256 values, according to each schema:

```
upper-triangular result:
35daf49284a6e7a47da4f5a69df82061d2d44536f7270978e3e8fefffeb47e8b
upper-triangular certificate:
c4b72e1179a80fddd0d34d1e1d650402966350e3d0315d144d4c586df371c322
Stage-3G-v2 result:
52d2c4df0cea7b6d98d898669e45ef54bfed1799965b2cff92161e84bd78ce13
Stage-3G-v2 certificate:
efad1bceffcf44790a2c753d140b3f183faeb1a004b453dcd4de5f7c7b44489b
Stage-3H-v2 result:
c0c5b854236f8403dacbb0037c5c409ad5f980364571bdf60fa9981a2e287408
Stage-3H-v2 certificate:
07ff0895fbe439939afd851cf136d6c9677ecb9dbb3e2acb80e5d8598c14a132
Stage-3I-v2 result:
5fdb1a843070ceb5887f7384431f2414989afc3cb741abc7f19ed44a333d4970
Stage-3I-v2 certificate:
fa82997b4b685ec2791be4d1efa82a0b6e5ba3b358d3d18bfcdd295c55bc7a24
Stage-4L result:
```

672f92c7c456a54f39afab7d2a5f92b783311cc0ee5341a4d2e72a588039017e
Stage-4L certificate:
b27631b62d437bf12431d751d4dad79570a03fa9a66fc0115d90f49768e058fc
Stage-4O result:
dc0e3951cb529dbdca384ff548ab0d7cd7786fe02573741e80e9c945452b2a23
Stage-4O contract:
95a204d942eeadd13868700b7353a8eca19b751a4d86f4080212424fe5880c97
Stage-4P result:
860a51d51648919f74bd7bd4e8230a629f7864b2bdcccf490aab5ff9e8e6b542
Stage-4P design:
9f40702bd413aa65da12973cd4fcc9b62a68b41355c490db3a192241b3fdaeb8
Stage-4Q result:
e4481bca2d021517073216dab15ee91c43cf301822b15e337c1b5061e9aaf49a
Stage-4Q pilot:
475e36d7ff956da5750607d30a8dc7dca185473aaca00c7ea7a6d5cd1dcd3157
Stage-4R result:
4e68835bc3ba5fd44432d98a3b6b1d41506533d66f3353cd500df3e95da76418
Stage-4R theorem:
56f7291b08052c999a2b89a96c41b2319c1d190bd9a1bddda7a7f52488021e16
Stage-4S-A result:
b552c5c6fc8afce53ed047ad8264a9d428351d9f031dc566af60969307a1d91f
Stage-4S-A certificate:
4fa88882ec4de4b50d3cda93a6b74d65796ea3138e939c6cb0e26ffff644c70c
Stage-4S-B result:
28e7116d855ec1a7e2b169356dce4d4dfb130069b35616a5ed6416f6634522e5
Stage-4S-B bridge:
095c947e19259b0d8a081738aad27a3860e2addf0070f63d1d9f070eb6bbf175
Stage-4S-C result:
fde3e1d6fc8d55dbaf4f33ef4098e335f3c4febe39c2b9ac78d327ad688fbc70
Stage-4S-C artifact:
e374068f4c6dc1ea2c8c40ddac64c21453efb93ec98d8765a13f890caa444770
Stage-4T result:
6998c1ac89440f180ebe753d1aa41c36c8a849183794fdc38c52f2d8d54e94e1
Stage-4T artifact:
c08107e05bb3dc293fb94d41f9d9a20c66b26983459bd885127e70a1d34061a9
Stage-4U result:
5217451e8ea58c42c1ab6066cf953685788c4f3bf1197c206e1e21e624e4808f
Stage-4U certificate:
65d5cf77166c3846a5dd0ffdd58ebe21e3d520ee1e9f3d33d97280b266a6d58f
Stage-4W result:
4877002a8be5347425f193d4ca239e5532e35551bee8c61de40649467fd2909e
Stage-4W artifact:
2fc75c41bc0bb7e00a17320f7de8cb7a112395a67b0c31b62e879ec1fad804c0
Stage-5F result:
26acd6d9421a8bf60d5bb96fe4c68918b39f89a443973dd728fed17ccf48652b
Stage-5F certificate:
e38f001e372372ef6186f16339d72c32302ad868441ef7eafbd0f533eb7c793b
Stage-5G-a result:

```
56e847fc804ced75e6c2fbf09ccbec1bdeabf505638e093c0939c2f2e584dd8c
Stage-5G-a certificate:
4bd0671fa9abb155f8ddd75eb7a4d26ce4ebbc783c073420784bdf54a1dbbcfa
Stage-5G-b result:
a16e9159d462c6b8f58851c2181147940f27e1a404ee4c2fbeb8999440cf8b64
Stage-5G-b certificate:
9fd06db88ce1743a27d27bef86e2d16891c92c39f8adf7e60ae3d4d00f7ef813
Stage-6A result:
f199cdc2cba603ef9c4a0cb7e5e5383a85f749b05dfa1f99d83245580697caac
Stage-6A certificate:
a4afa5902d95fcbfa50477bae7c30b6998097fd823eab2ea3eee9c26cad0f144
```

The delayed-network parent/child chain passed $28 + 41 = 69$ tests, including temporary-repository warm-cache attacks that refresh the visible parent and child hashes after mutating a live parent source. The Stage-3H-v2 release passed its 18-test suite, including a fresh Stage-3G parent replay. Stage 3I passed all 18 tests, including a public cache-cleared fresh numerical replay. Stage 4L passed all 31 tests, including a genuine fresh numerical replay. Its hostile and regression checks reject separate norming of the two rank-one corrections, omission of a delay word or output row, center-period-only coverage, circular use of K_s, inward rounding in the theorem note, incorrect recovery-inclusive rectangle arithmetic, and every promotion to first return, nonlinear tube, stable graph, crossing, onset, routing, or network safety. Stage 4O freezes the analytic event-Hessian dependency order and, jointly with its parents, passed 102 tests. The fail-closed Stage-4R-v2 theorem passed 34 direct tests, a fresh replay, and a 66-test joint regression with Stage 4O. Stage 4P passed all 33 tests, including a fresh replay and hostile rejection of missing physical normalization or proof promotion. Stage 4Q passed all 24 tests and a fresh $n_{\text{step}} = 120, 180, 240$ replay; all nineteen theorem flags remain false. The repaired Stage–4S-A full/reduced-history event tube passed a 147-test parent-chain regression. Stage–4S-B's finite-sampling no-go and atomic bridge passed jointly with the replayed 4O–4Q chain in 117 tests; none of the six Hessian flags was promoted. Stage–4S-C and Stage–4T passed an 80-test direct chain and a 145-test regression with their smoothing and terminal-row parents. Stage–4U passed 21 direct tests and a 166-test parent-chain/fresh-Stage–4L replay, for 187 passing tests in all, on its explicit 9×10^{-31} event tube. Stage–4V passed all 7 direct tests, including a fresh arithmetic replay, and a 190-test eight-parent regression with the duplicate replay deselected. Two independent frozen generations were byte-identical. Its result, certificate, and numerical-core SHA–256 values are respectively

```
e371d5c80b0d9dafa2e4a3164650f96dfb6e6ed7d0ed170db438479db42d8ad0
51336d8b7ae494e8923cf5c1714b0be7543bb0372ec3fd5410695e859e4c2d08
7cda699176730596d1cfc0c1445b14d5f52840aa8a64b8e18d373618bd76ec4e
```

An independent final replay additionally passed the Stage–4V suite and a 151-test parent chain. The exact-binary period denominator, true-delay phase drift, and the distinction between a failed preferred-ball diagnostic and a validated flow bound are part of the frozen certificate. Stage–4W passed all 18 direct tests, including a fresh replay and hostile checks that keep every numerical center block and residual unset. It proves the exact stable quotient norm, split-projective residual implication, and qualitative local selected-map stable graph while retaining numerical ingress at 0/6 and every physical-crossing and onset flag as false. The Stage-5F release passed its 20-test suite and an independent cold validator. Its conditional graph-slope flag is kept distinct from every false graph, root, onset, routing, capture, and network-safety flag. The Stage-5G-a endpoint release passed its full 26-test suite: 25 non-recompute tests and one fresh-recompute test of its proof-bearing numeric core. Stage 5G-b passed its 22-test suite and a separate fresh-interpreter replay. The latter recomputes its exact-rational derivative, cone-intersection, radius, and conditional-slope arithmetic while normally validating Stages 5D, 5E, 5F, and 5G-a. Both releases keep graph, stable-gap, crossing, onset, routing, capture, and safety flags false. Stage 6A passed all 14 tests, including a fresh directed recomputation. Its local event, no-earlier-hit, nonlinear return, and complete-history tube flags are true, while $J = 0.32$ entry, outer capture, two-sided routing, onset, and safety flags remain false.

The wide pulse-family certificate, for example, was regenerated at 192-bit precision over all 1152 cells; the regenerated certificate agreed with the frozen one byte for byte. The Route–C event certificate was independently recomputed and reproduced its certificate hash; a hostile test rejects the obsolete Fourier candidate voltage as an exact section level. The event-aligned first-derivative certificate was regenerated twice, closes every first-variation cell, and has hostile tests for omission of the translation term or illicit differentiation of the Stage-5B remainder. The Route–C adjoint certificate likewise has an independent replay of the Fourier tail and normalization. A separate source-bound control contract binds the event certificate by both result and certificate hashes and exposes only its six event-side conclusions; hostile tests reject any stable-sheet-to-onset promotion. The repository records exact commands, arithmetic scope, and claim ledgers. These checks make the computation auditable, but they do not enlarge the scope of the theorem encoded by the certificate.

The outer-resolvent audit illustrates why byte reproducibility alone is not a proof. A later endpoint check found that the nominal 12,000-patch cover contained only one of two physical terminal-clipped cells in each δ-strip. The affected Stage-3G result and every dependent Stage-3H claim were withdrawn publicly. Their v2 replacements use the longer equation continuation on 730 ordinary and 40 clipped rectangles, hence 12,320 local patches, and have been independently replayed. No withdrawn v1 residual, Green, or output-row number is used; the Stage-3G/3H-v2 parents themselves make no E_v, E_w, C^0-contraction, or attraction claim. Their separate Stage-3I child below closes the reduced-history linear contraction; Stage 6A is the separate nonlinear local-tube child.

5.4 Outer linear transfer and a microscopic nonlinear tube

The phase-fixed binary shadow of the outer return has

$$\|P\|_\infty \le 0.126907895, \qquad \left\|P^2\right\|_\infty \le 0.002761,$$

whereas the fixed-time and rank-one phase terms are separately of order 2.58. The small return is therefore a signed cancellation, not a positive-kernel estimate. A continuous Volterra shard has been validated through the first nonzero delayed-forcing boundary, but a direct global tiling would require approximately 1.05×10^6 coarse tasks before adaptive children and second-delay faces. The local uncorrected kernel masses are too large to serve as the transfer errors E_v, E_w.

The exact delay support gives a finite alternative. Since

$$3\tau_0 - T > 0.2283989174,$$

all delay words of length at least three vanish on the return window. The global continuous kernel is therefore an exact sum of only 14 history word integrals and 7 recovery-scalar word integrals, each of ordered dimension at most two. This reduces the representation count by a factor greater than 49 914. The finite-word compression is rigorous and fixes the signed order of phase subtraction and cross-word summation before total variation; low-rank templates used only to design the quadrature remain diagnostic.

The ordered two-simplex terms have now been reduced exactly to eight one-dimensional primitives $F, G, H_0, H_1, L_{00}, L_{01}, L_{10}, L_{11}$. A degree-24, 1024-cell relative-residual certificate closes F and G on the exact 10^{-8} orbit/period ball:

$$\eta \leq 0.005026008, \qquad \left\| \widehat{F}^{-1} F - \mathrm{I} \right\|_\infty \leq 0.005038659, \qquad \left\| G\widehat{F} - \mathrm{I} \right\|_\infty \leq 0.005064176. \tag{190}$$

The determinant of every polynomial chart stays at least 1.29906×10^{-5} from zero. This is the first global exact-orbit transfer step, but propagating the F/G error through H and L with separate absolute norms gives unusable bounds 190.1 and 7.17×10^6. Those numbers are a rigorous rejection of a cancellation-blind triangular estimate, not estimates of the return operator.

The combined-row adjoint reduction now places every delay word, both history injections, and the phase subtraction inside one advanced row before total variation. Its directed instantaneous calculation proves

$$\int \|G_{\mathrm{inst}}\|_{\mathrm{phase}} \leq 14977.452934, \qquad \|G_{\mathrm{inst}}\|_{\mathrm{bdry}} \leq 66623.270373,$$

leaving positive delayed allowances 45022.547066 and 3376.729627 under the declared full targets. It also proves full coefficient variation at most 1.633190×10^{-6} and phase-ratio transfer errors at most 1.360127×10^{-7} in voltage and 2.904969×10^{-8} in recovery. These are architecture and instantaneous bounds, not the delayed signed-density residual.

The corrected global tensor enclosure uses

$$S(\delta, \ell) = R(T - \delta, T - \delta - \ell),$$

on a degree-$(10, 24)$, cutoff-128 Fourier–Taylor cover with 4-by-4 local patches. The endpoint identity

$$\frac{T}{h} = 47.5914275778\ldots$$

shows that the physical domain contains both lag cells $46 -$ delta-cell and $47 -$ delta-cell at its terminal edge. The first implementation included only the former. Its continuation length $48h - T$ therefore covered the first clipped layer but not the second, which requires continuation through $49h - T$. The v2 construction performs that continuation and validates all 12,320 local patches. Stage 3G gives strict exact-guide Green and boundary bounds 71.896340 and 261.764130, respectively. Stage 3H then keeps the phase subtraction inside each output-specific row and proves center-guide full-row bounds

$$\|p_v\| \leq 41.482447, \qquad \|p_w\| \leq 1.177738. \tag{191}$$

These are not the transfer errors E_v, E_w. Stage 3I subsequently sums both delayed injection branches inside each continuous center-density cell before taking absolute values, covers all 25,600 rectangles and all lag-chart seams, and adds each candidate-to-guide and guide-to-exact cost once. Its auxiliary 0.01 candidate-row excess target fails, but that target is not the contraction gate. The certified exact row budgets are

$$Q_v^{\mathrm{exact}} \leq 0.55051563144094195 < 1, \qquad Q_w^{\mathrm{exact}} \leq 0.028280815376548179 < 1. \tag{192}$$

They prove contraction of the phase-fixed linear return on $C([-\tau_1, 0], \mathbb{R})_v \times \mathbb{R}_{w(0)}$, restricted by $h_v(0) = 0$, for arbitrary continuous histories in the reduced norm. These row budgets are the linear parent for the following local nonlinear result.

Theorem 5.1 (Microscopic nonlinear outer local return tube)**.** *Set*

$$Y_o = C([-\tau_1, 0], \mathbb{R})_v \times \mathbb{R}_{w(0)}, \qquad \Sigma_o = \{h \in Y_o : h_v(0) = 0\}.$$

In coordinates centered at the exact outer phase-zero history, the ball $\overline{B}_{\Sigma_o}(0, r_6)$*, with*

$$r_6 = 10^{-335}, \qquad R_6 = 10^{-3},$$

has a unique next positive local-section event, no earlier hit of the same local section disk, and a C^2 *return map* P_o*. The map is a strict contraction with*

$$\Lambda_6 := \mathrm{Lip}(P_o) \leq 0.561838642158326 < 0.561839. \tag{193}$$

For every compatible complete history above this reduced ball whose old recovery trace is within radius R_6*, the intervening complete-history flow sweep remains in the full radius-*R_6 *tube and returns to it. Consequently this forward tube is locally attracting under successive returns.*

The source-bound proof covers the normalized phase circle by 256 cells. On the 254 middle cells, outward Fourier–Taylor–Bernstein arithmetic gives exact-orbit current-state separation greater than 0.0912678; on the two wrap cells the exact current-voltage speed exceeds 0.8262423. These inequalities provide the local event chart and exclude an earlier hit of that local disk. The same-tube second-variation and event-time estimates give

$$C_P \leq 1.132302 \times 10^{333}, \qquad Q_v^{\mathrm{exact}} + C_P r_6 \leq 0.561838642158326 < 1.$$

The flow-growth estimate keeps the radius-r_6 reduced ball inside the radius-R_6 complete-history tube, and the returned history window lies strictly after the initial time. This closes complete-history invariance without using an effective inner stable graph.

The directed $J = 0.32$ parent supplies only the ambient complete-history bound

$$d_{.32} \leq d_{.32}^{\mathrm{up}} := 2.637078616900037 \times 10^{-5};$$

it does not place that history on the exact outer section. Even the optimistic comparison $d_{.32} < r_6$ fails by about 330 orders of magnitude. Thus the first failed biological gate is *ambient-to-section domain containment*, before any capture inference. At radius equal to the directed attachment bound, contraction would require

$$C_P < \frac{1 - Q_v^{\mathrm{exact}}}{d_{.32}^{\mathrm{up}}} = 17044.784545\ldots < 17044.785. \tag{194}$$

Theorem 5.1 therefore proves neither $J = 0.32$ entry nor outer-side attachment, capture, routing, or onset.

5.5 Outlook

The finite-network root theorem and the physical-pulse program share a collective–transverse architecture but solve different problems. The former shows that a simple selected history root survives arbitrary finite network size under a uniform mixing hypothesis. The latter seeks a model-specific selected stable-sheet threshold and its two basin attachments. The endpoint functional signs, the full radius-47/5000 stable-coordinate cone, and the discrete inner stable-power input are now proved. The general eventual-smoothing theorem, exact moving-event Hessian, microscopic arbitrary-history event tube, direct identity $DQ_Y(Y_*) = A^2$, fixed splitting and rates, exact split quotient and bimeasure residual theorem, qualitative local selected-map stable graph, six-block pilot, and conditional graph arithmetic now reduce the effective inner problem to one explicit certificate chain: enlargement of the selected near-two-period return to the preferred anisotropic graph ball, graph-transform return-domain containment, all six directed continuous-history Hessian blocks in the wide box equation (130), and either the direct-return stable-set hypotheses of theorem 2.6 or a nested-domain identity $Q = P^2$. The residual theorem identifies the exact split-projective norm in which the missing six-block computation must close; its present numerical ingress is 0/6. Once that chain closes, the already proved normalization adapter yields the graph-adjusted endpoint signs, strict slope, and unique selected crossing without a new graph-arithmetic idea. Remaining biological gates are entry of the $J = 0.32$ history into an outer return domain, outer-side attachment, and two-sided routing. The next outer enlargement target (Stage 6B) is a signed intermediate-flow/second-variation kernel that keeps the event-time correction and kernel cancellation together before total variation, with $r \geq 2.64 \times 10^{-5}$ and $C_P < 17044.785$. These are targets, not conclusions of Stage 6A. Once the remaining gates are closed, the formulas in equations (181), (185) and (189) give a rigorous local inverse from controls to frequency, voltage amplitude, and a network-robust biological safety margin. Before routing is closed, the same formulas describe only a threshold-offset target coordinate and local network-gap side.

Diagnostic inner nonlinear-tube target. A source-bound Stage–4N feasibility pilot tests the preferred anisotropic ball $\|x_s\|_Y \leq 0.0097$, $|\widehat{x}_u| \leq 0.00025$. Two disclosed cancellation-blind scalar Gronwall constructions miss their own containment tests by directed lower values $11.971029\ldots$ and $32.257662\ldots$ decimal orders, reported diagnostically as 11.9710 and 32.2577. This does not lower-bound the true nonlinear flow. The pilot identifies one sufficient future signed event-aligned target $K_{\mathrm{ret}} < 188.9122238810816$ (numerically 188.9122238811), after all stable/unstable, event, and translation correlations are formed before the norm. Its frozen result SHA-256 is

`5e7214a2f5ba8ca22649c677a1d054b32342b5cc25966bd8e1da7600c605f1de`

The later split analysis proves that this ambient scalar target is not necessary for graph closure. In particular, the Stage–4Q signed near-two-period pilot has ambient heuristic envelope 339.1671, so it fails that scalar route, while each of its six projected blocks lies far inside the simultaneous graph box equation (130). Stage–4Q is still a nonrigorous finite-section diagnostic: its result SHA-256 is

`e4481bca2d021517073216dab15ee91c43cf301822b15e337c1b5061e9aaf49a`

and all nineteen theorem flags remain false. A C^2 selected nonlinear hit and a qualitative local stable graph for its selected coordinate map exist. The scalar event tube has scale 9×10^{-31}, and the orbit-aware weighted-energy theorem enlarges its certified closed scale to 1.2×10^{-8}. This remains about eight decimal orders below the preferred ball, while preferred-ball event/return containment, the six numerical continuous-history Hessian blocks, an effective stable graph and its periodic-orbit stable-set identification, crossing, onset, routing, capture, and safety flags remain false.

The resulting theorem will not say that every delayed network has a canard or that every pulse threshold is a canard. It will say something both stronger and more precise: within a dimension-uniform finite-network class, a declared complete-history canard root has controlled structural response, and, once the remaining model-specific gates close, a separately validated physical pulse crosses a history-space stable sheet whose two routed sides form a basin boundary and hence an independent control coordinate.

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Data and code availability. Source-bound certificates and their reproduction code are maintained at `https://github.com/h-lu/canard-aware-network-control`.

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